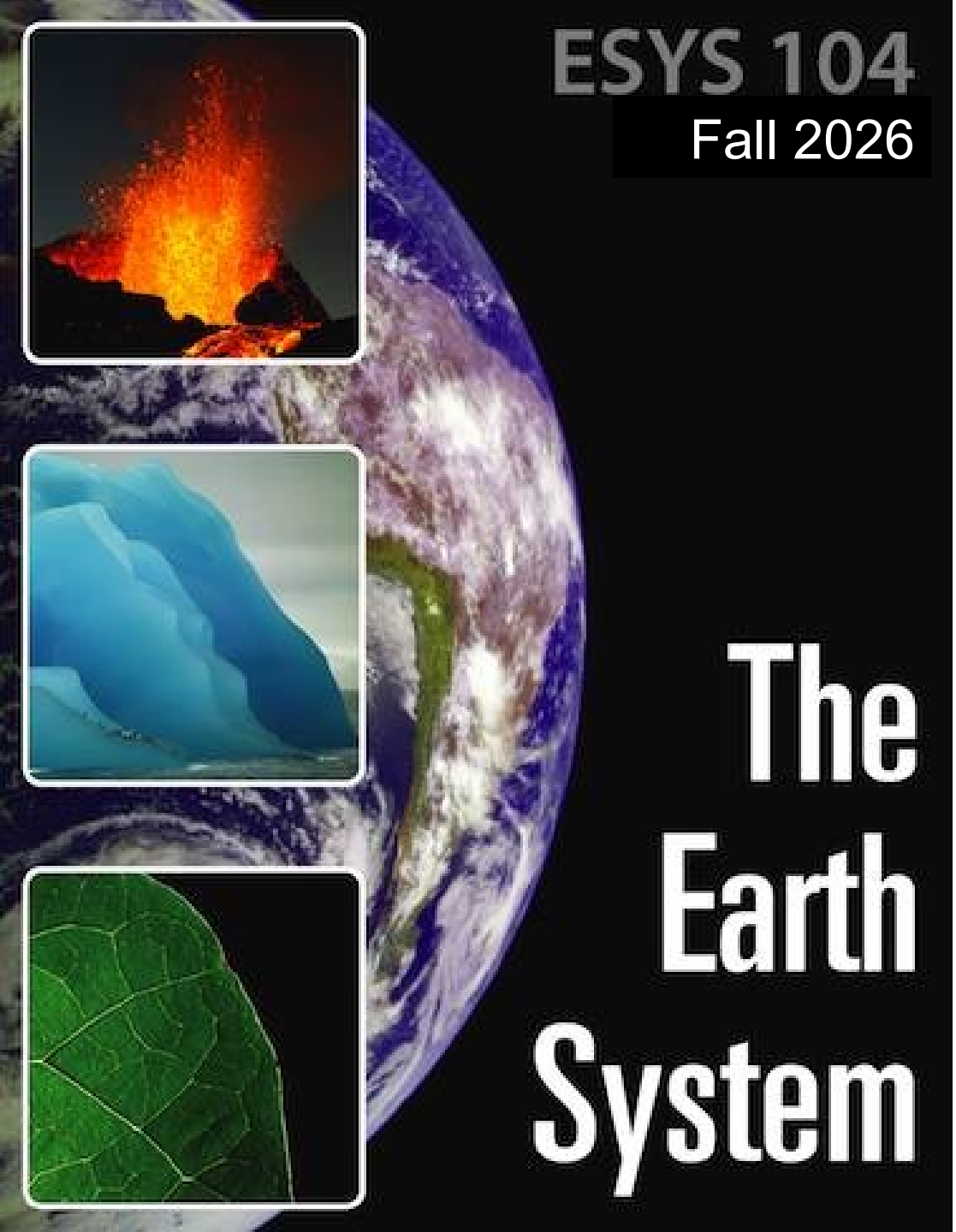
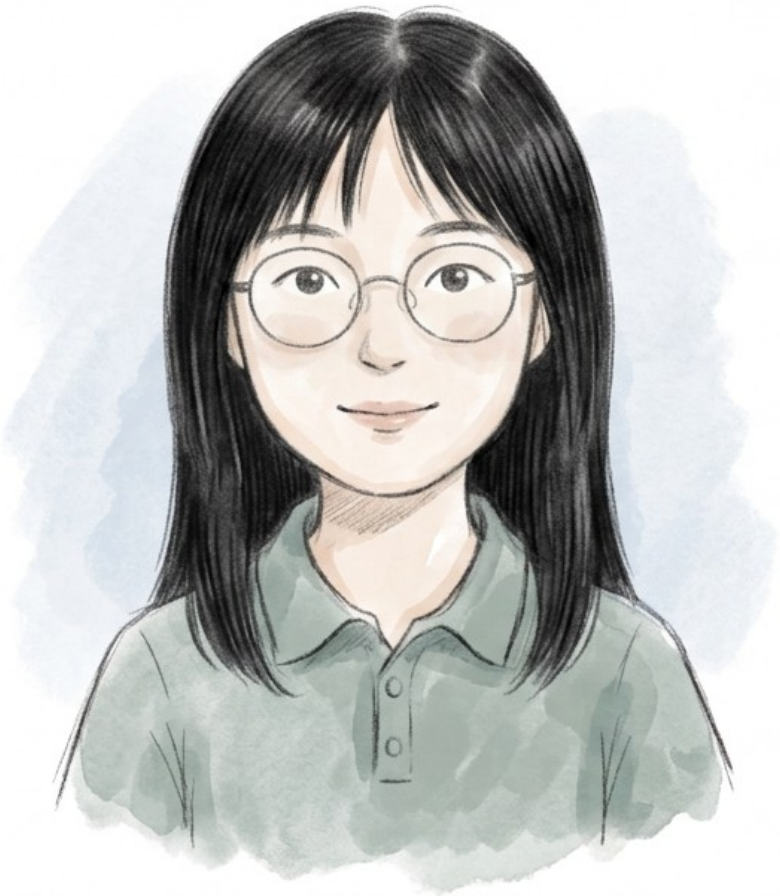


Lect #2: The Solar System and Earth Materials



Yan Su

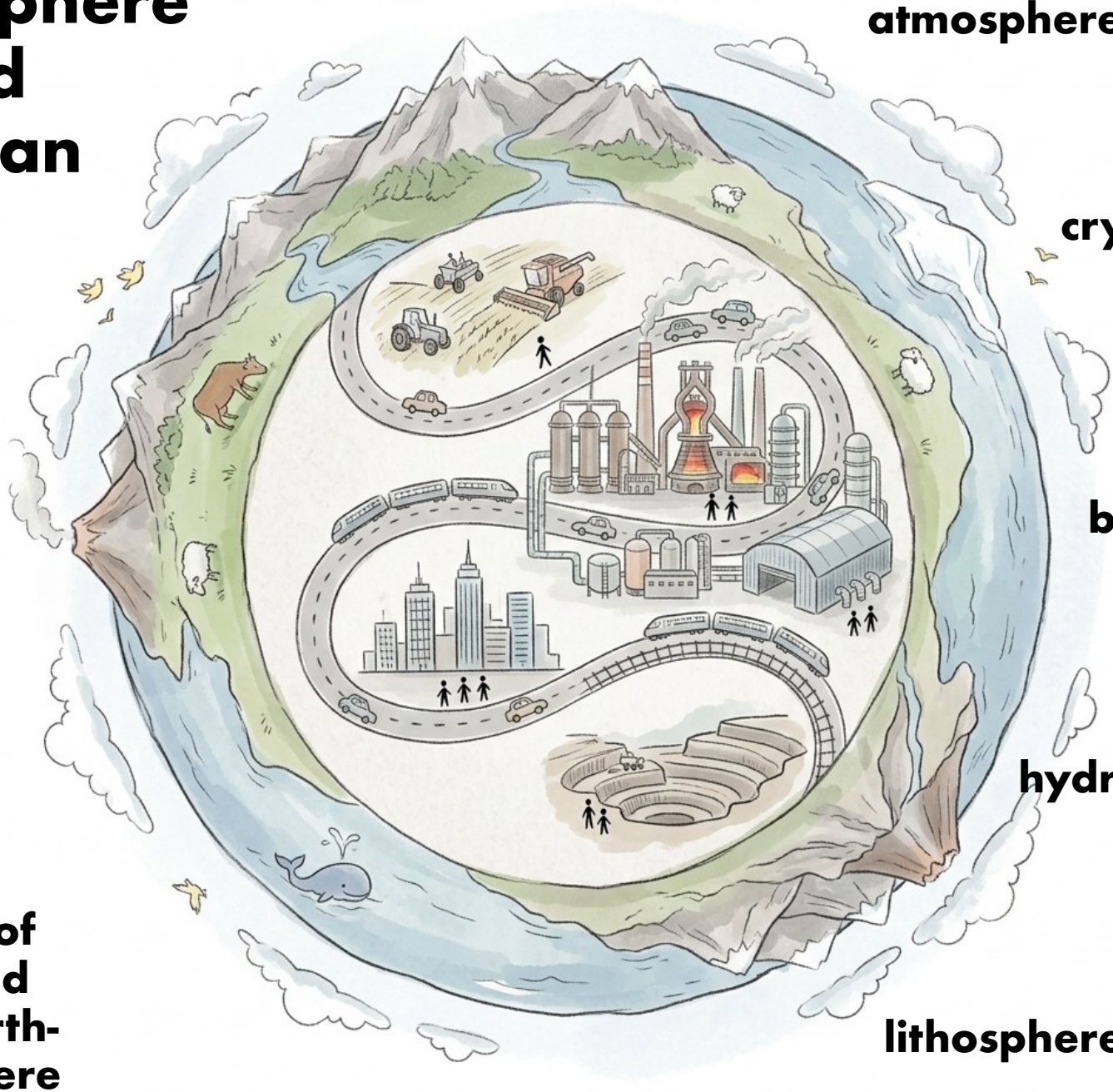
Master Student from Department of Earth and Planetary Sciences

Contact:
yan.su2@mail.mcgill.ca
yan.su2@mcgill.ca

Book an office hour with me by email or simply drop by my office FDA 209 anytime!

"Everything is connected to everything else!"

Technosphere and Human



A model of integrated human-Earth-technosphere

Jordan Thakar

- **Environmental microbiology**
- **Bioremediation**
- **Extremophiles**
- **Microbial Evolution**





- **Thesis:** Can microbes clean up oil spills in Arctic seawater and sea ice?
- **General:** How do microbial communities interact to shape ecosystem processes?





Reading Materials

Lecture 2: Earth Materials (today)
https://wiki.seg.org/wiki/Layers_of_the_Earth

https://opentextbc.ca/physicalgeology2ed/chapter/1-4-minerals-and-rocks/

https://opentextbc.ca/physicalgeology2ed/chapter/3-1-the-rock-cycle/

Look over: Chapter 2 Minerals

Lecture 3: Plate Tectonics (next Wednesday)
https://opentextbc.ca/physicalgeology2ed/chapter/1-5-fundamentals-of-plate-tectonics/

Look over: Chapter 10 Plate Tectonics

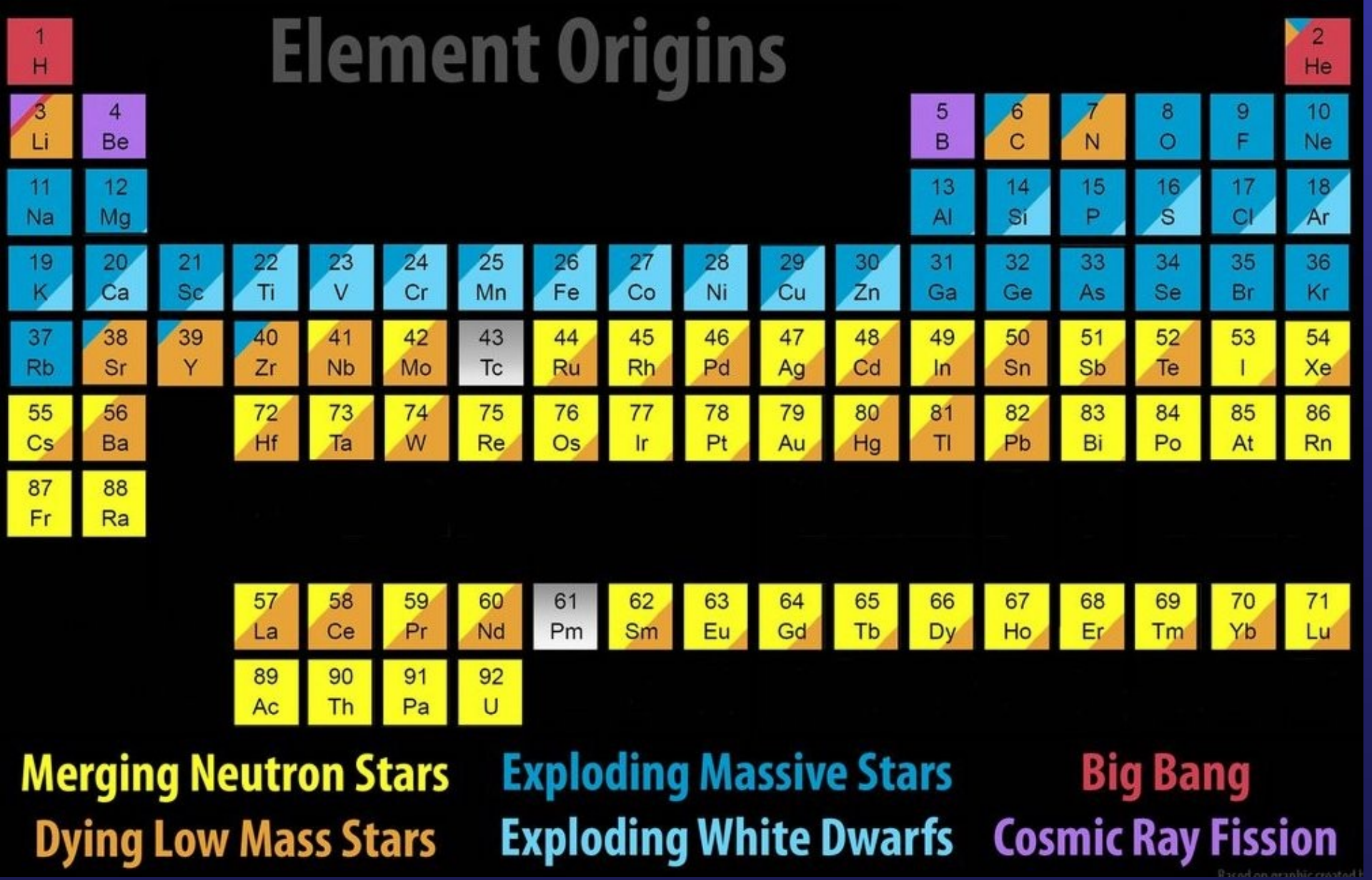
Physical Geology – 2nd Edition
By Steven Earle
https://opentextbc.ca/physicalgeology2ed/

Dust Lane, Spiral Galaxy



NGC 891, Kitt Peak, NOAO

That dust is elements (heavier than helium) produced in stars



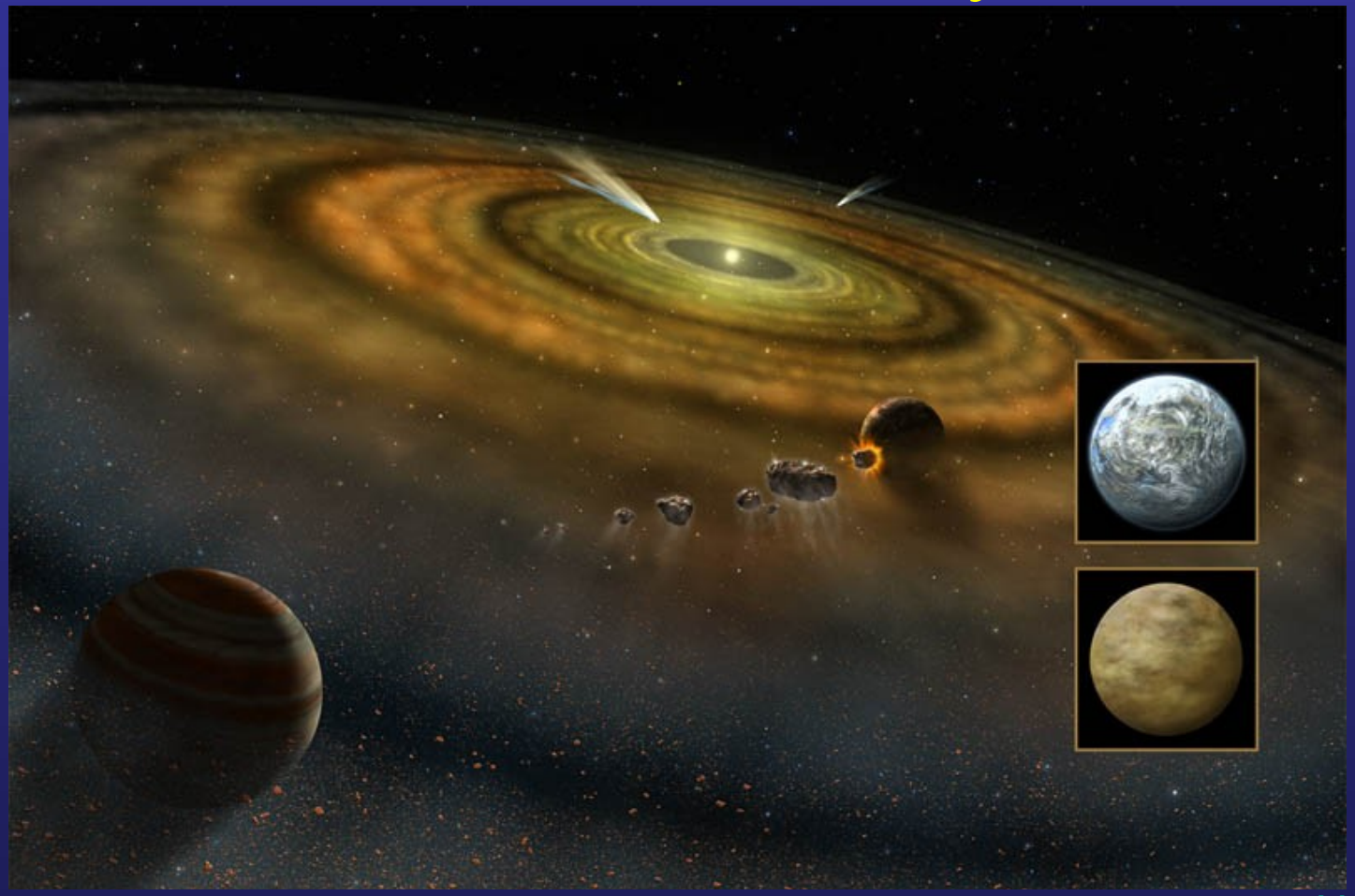
We didn't know the formation mechanism of the Yellow Elements until October of 2017!

McGill Connection



LtoR: post-docs Melania Nynka/John Roan with Daryl Haggard

Accretion of the Solar System



NASA

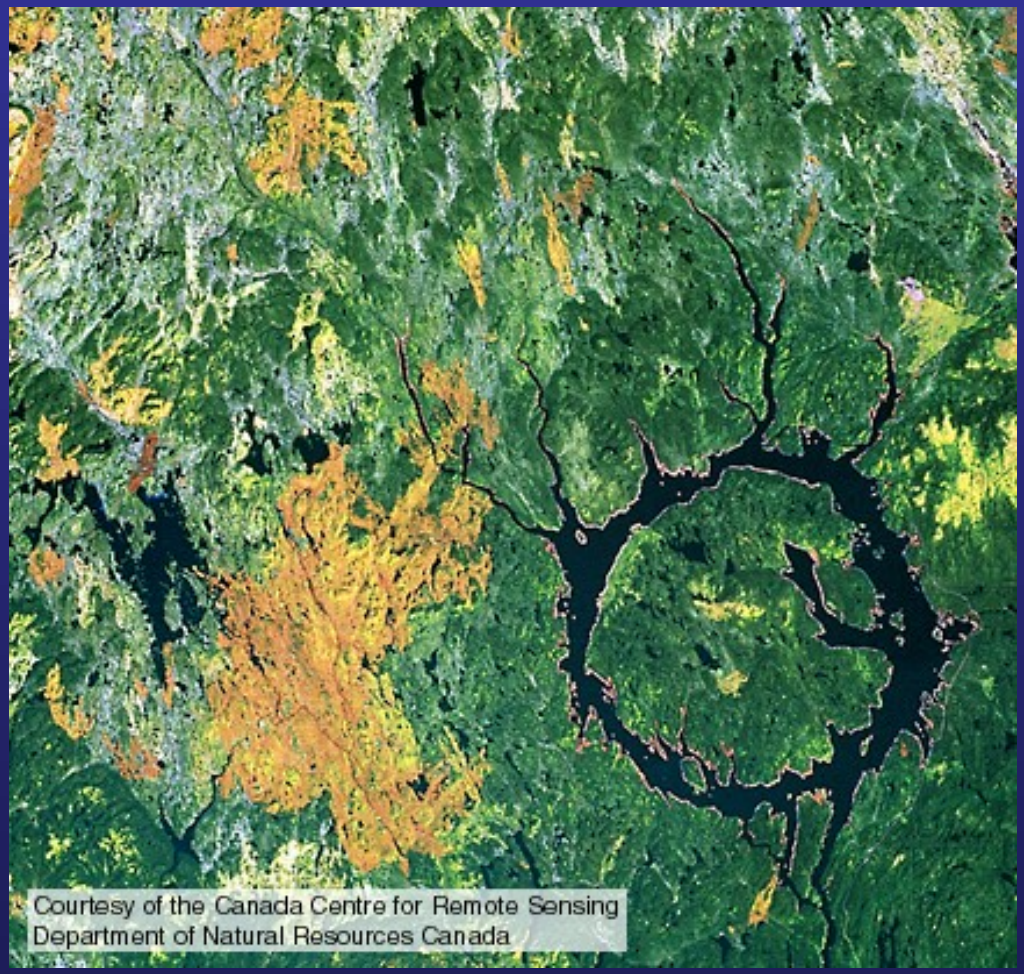
Gravity collected this interstellar material into a spinning accretionary disk, from which the planets formed. This occurred about 4.57 billion years ago.

A Giant Mars-sized Impact



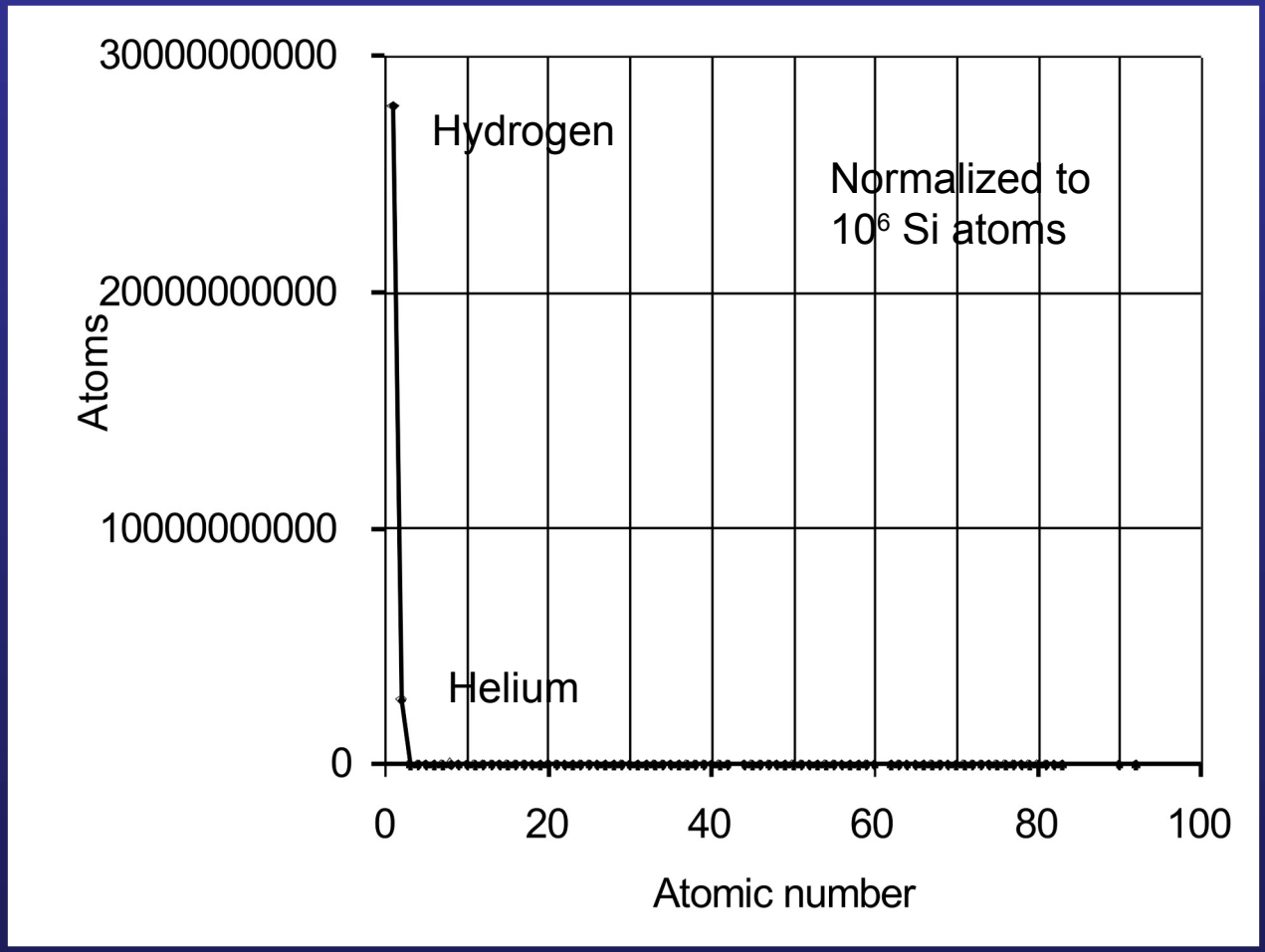
Large, Late Impacts were common as the Solar System was Cleared of Debris. One of these impact may have created the Earth's Moon

Scar of an ancient impact



Manicouagan Crater in Quebec formed about 210 million years ago from a meteorite impact

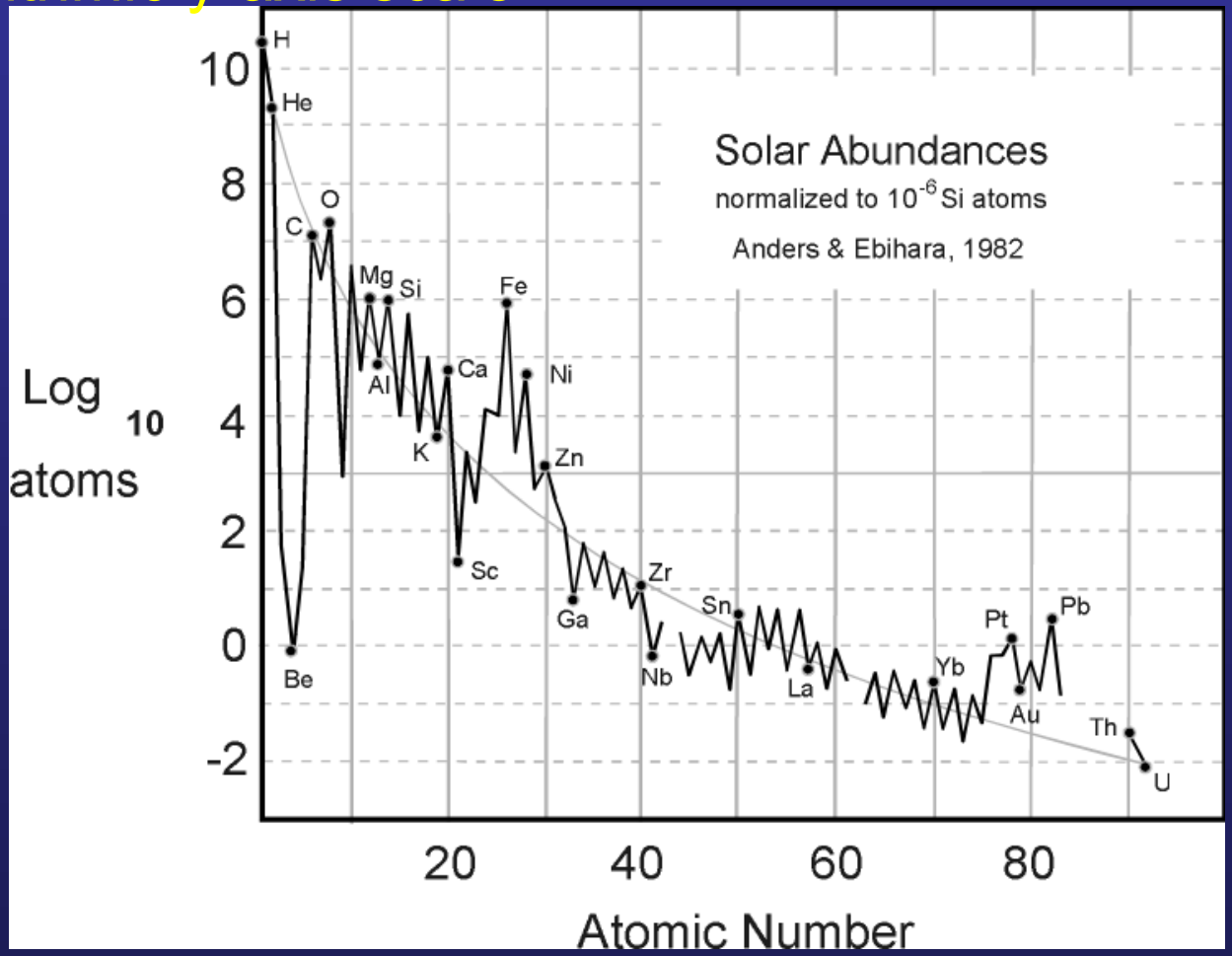
Solar System Elemental Abundances



Nothing but H and He and a few trace elements

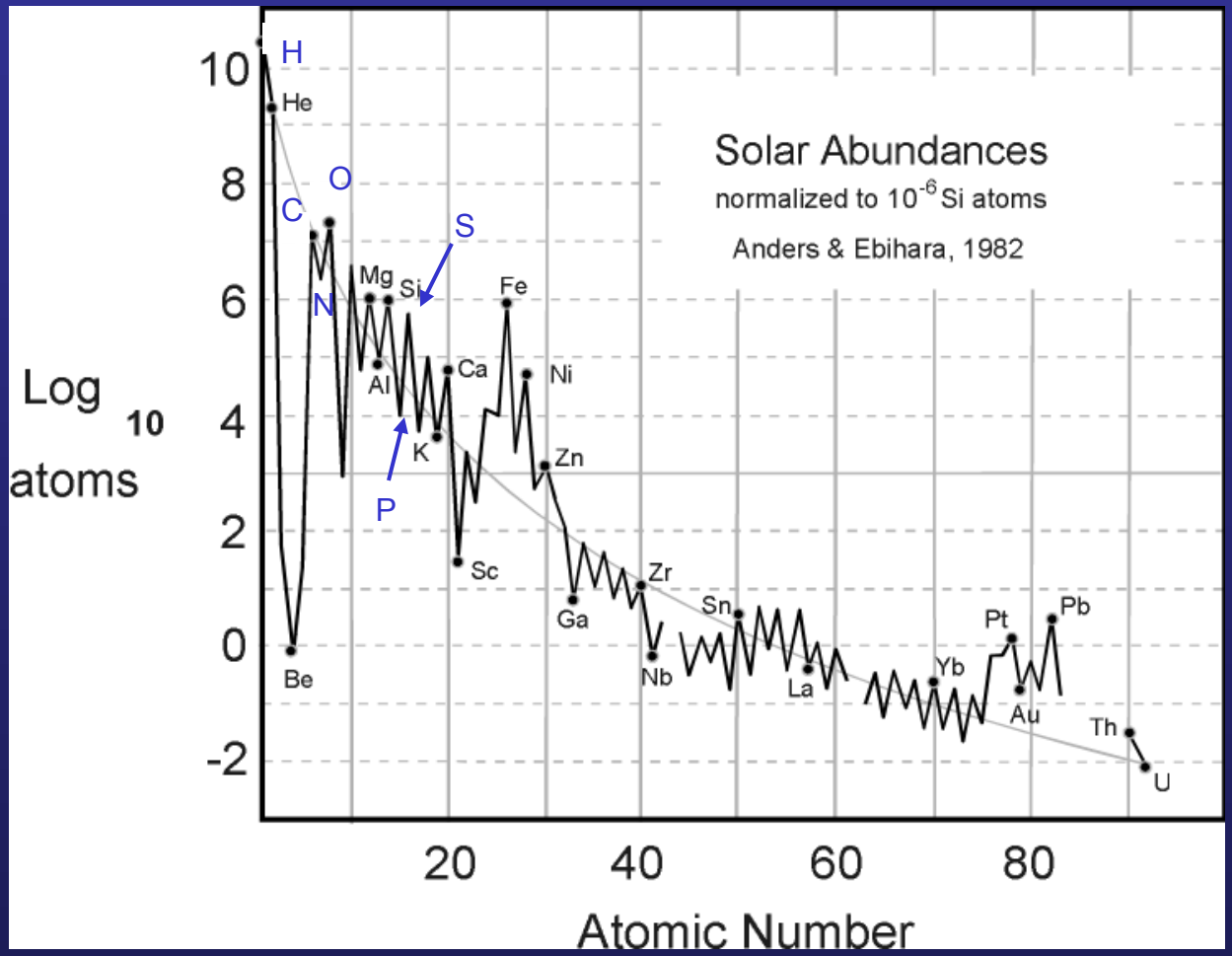
Elemental Abundances

Note logarithmic y-axis scale



Iron is the heaviest element that releases energy it is created (note the increased abundance)

Elements essential for life



Six essential elements:

Hydrogen, carbon, nitrogen, oxygen, sulfur, phosphorus

Elements essential for life

Glucose (a sugar) is a carbohydrate with the molecular formula $C_6H_{12}O_6$: each molecule of glucose is made up of 6 carbon atoms, 12 hydrogen atoms and 6 oxygen atoms.

Amino acids (which make proteins) additionally contain nitrogen and sulfur.

These and phosphorus are essential to nucleic acids (which make up DNA).

A number of other elements are essential trace elements for humans.

Six essential elements:

Hydrogen, carbon, nitrogen, oxygen, sulfur, phosphorus

Composition of the Earth

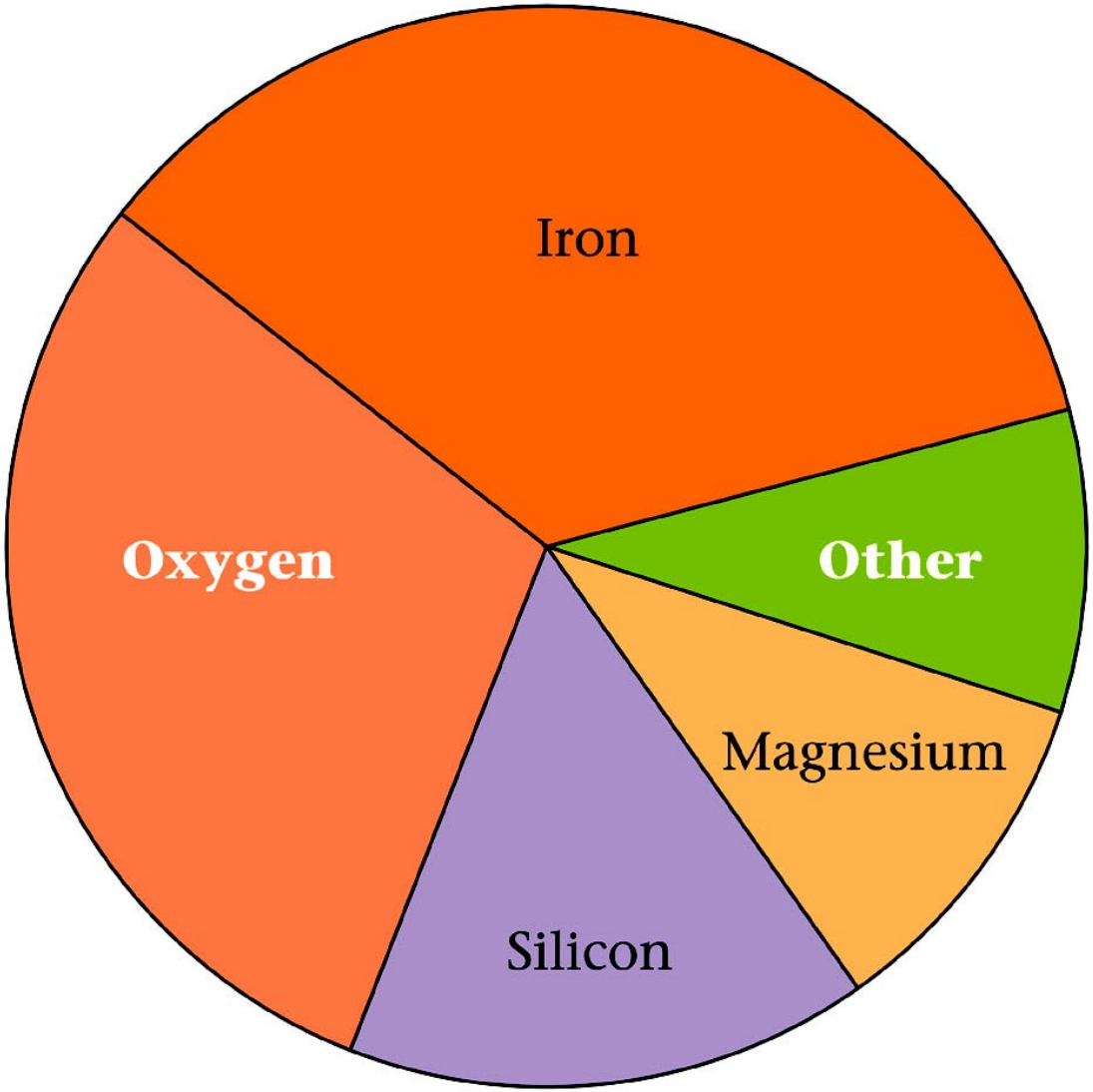
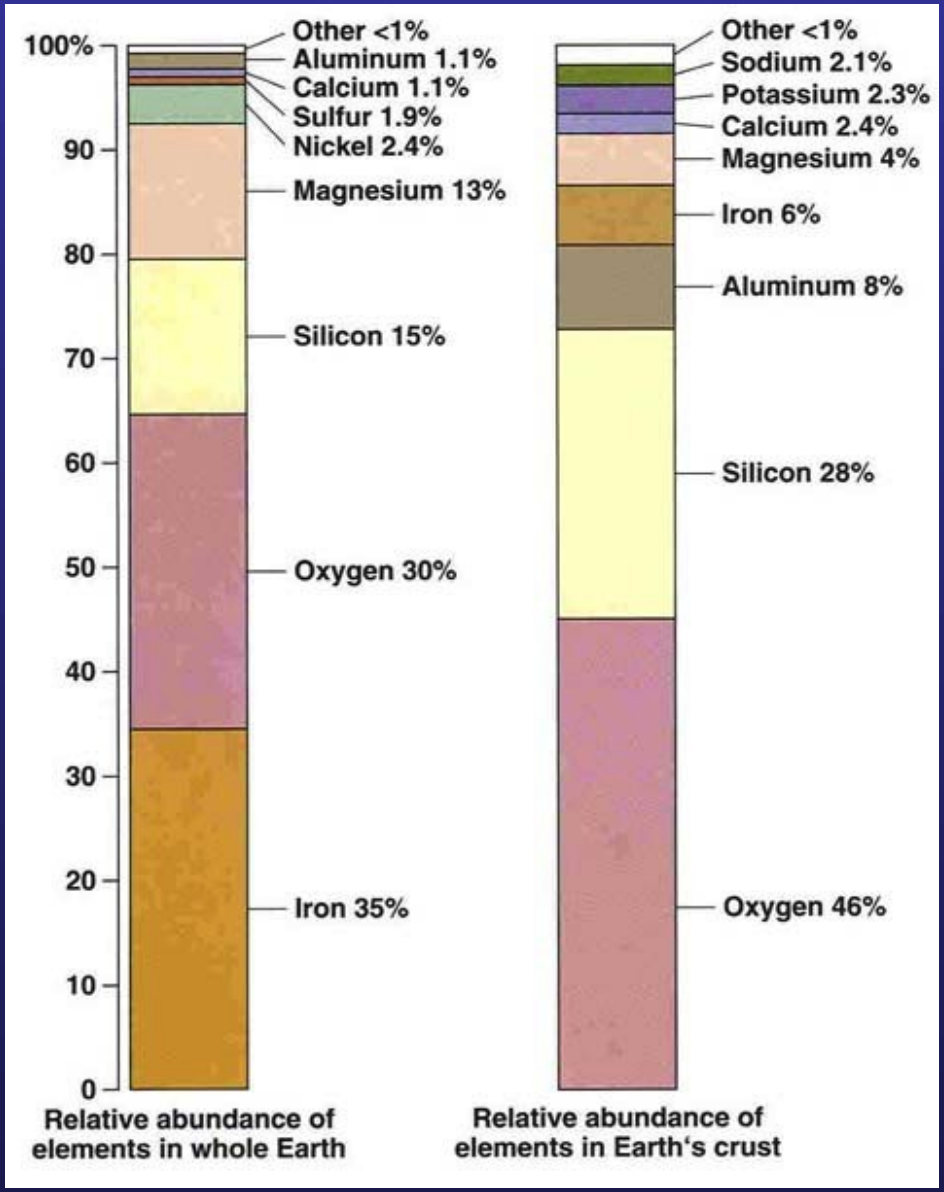


FIGURE 2.10

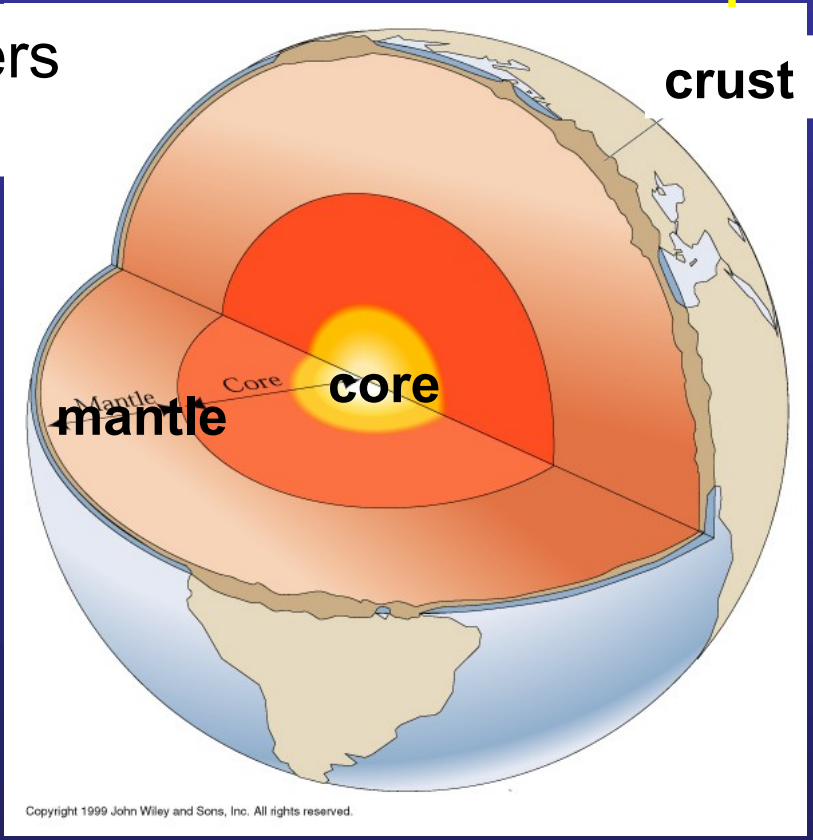
Earth: Portrait of a Planet, 2nd Edition
Copyright (c) W.W. Norton & Company

Composition



A differentiated planet

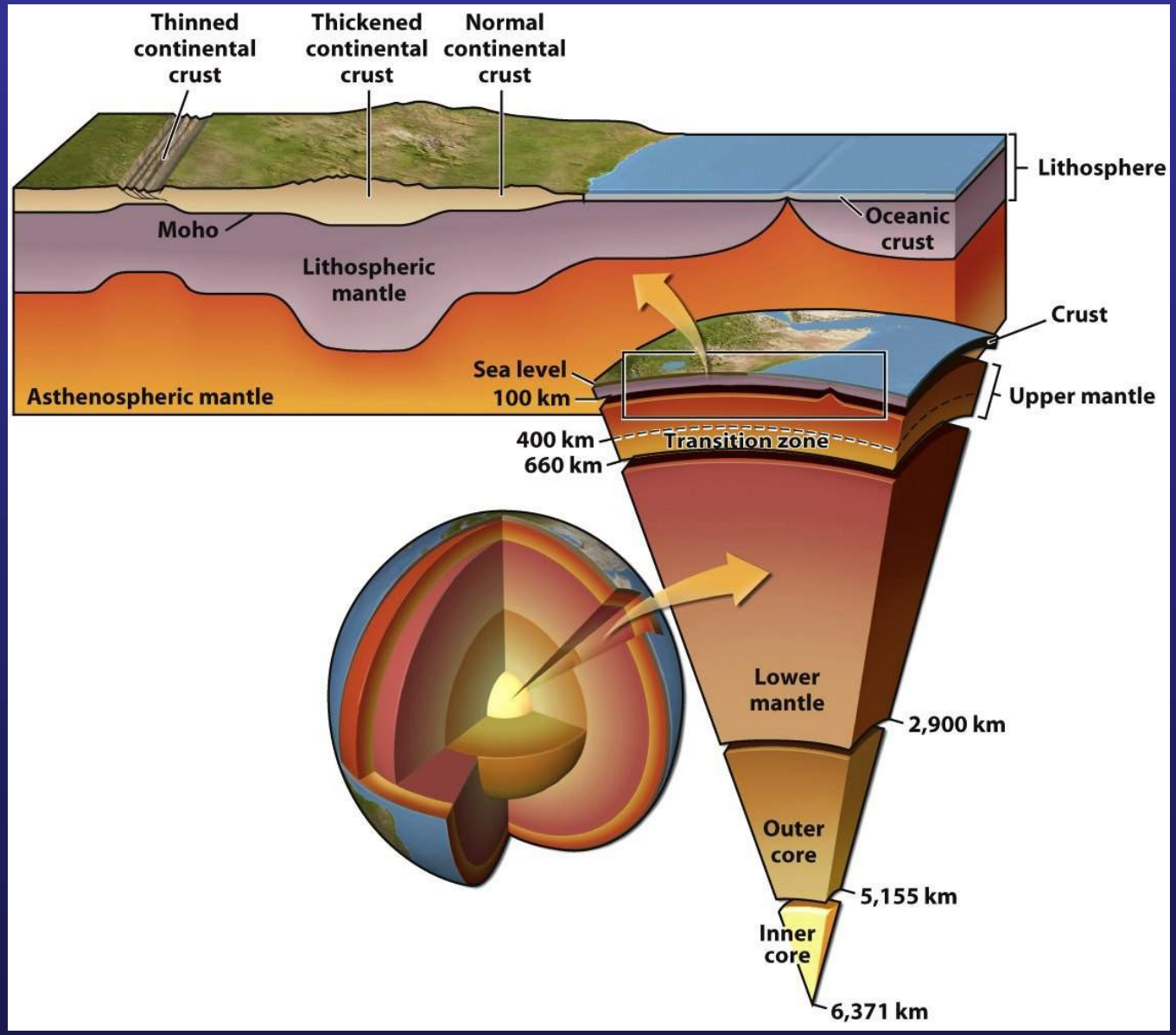
6371 kilometers to center



Driving at 100 km/h straight down, it would take 64 hours to get to the center

- During early **partial melting** the rocky inner planets **differentiated** into three layers of different chemical and physical compositions
- **core** (inner & outer) -- metallic
- **mantle** -- rocky, intermediate density
- **crust** -- rocky, low density **oceanic** and **continental**

Earth's Interior Layers



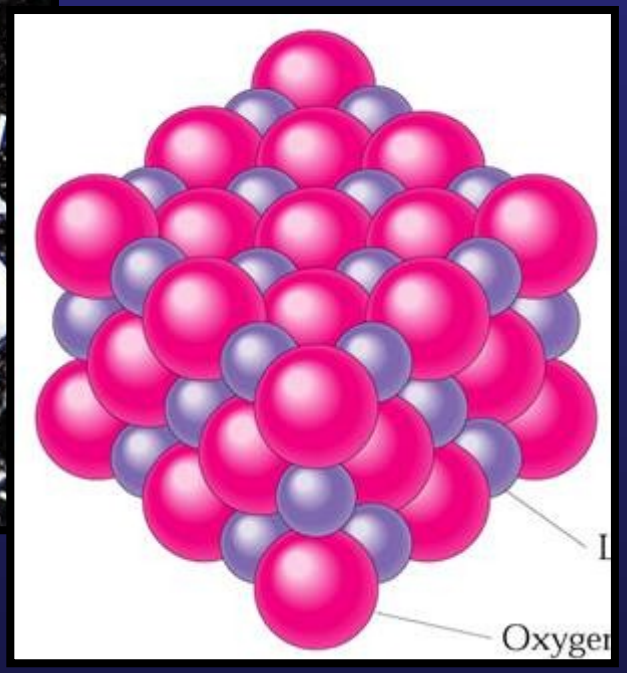
© W. W. Norton

Rocks, minerals, and atoms

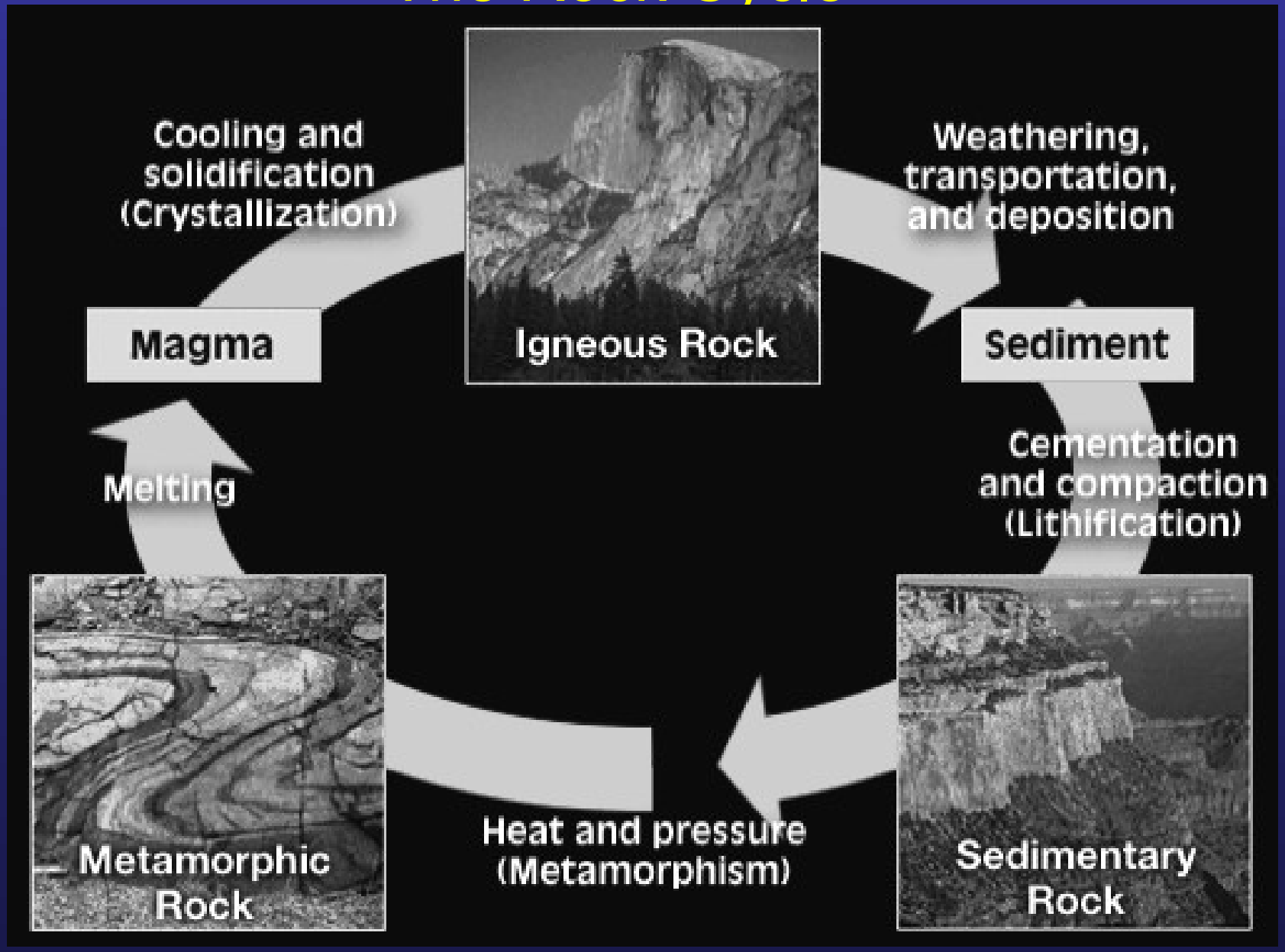




Rocks are made of minerals
are made of Atoms



The Rock Cycle



Igneous Rocks, some examples



Basalt





Granite



Sedimentary environments, a few examples

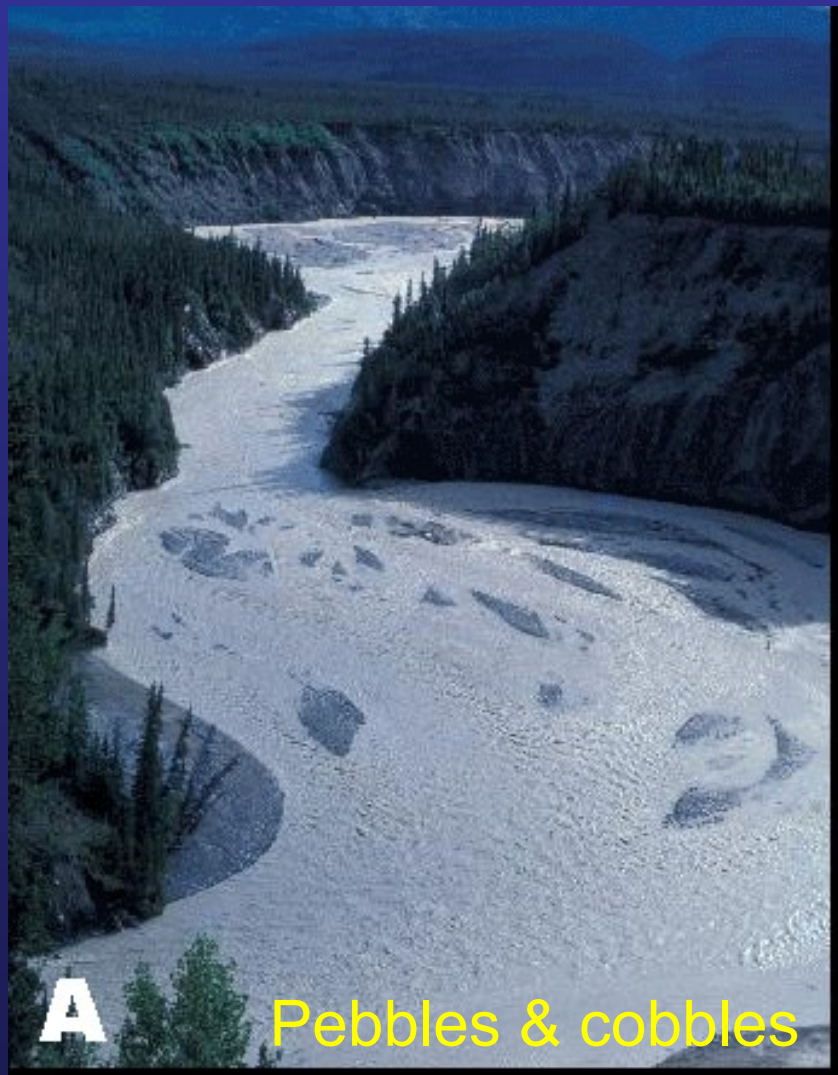


Sand

Modern and ancient beaches

Sandstone

Sedimentary environments, a few examples

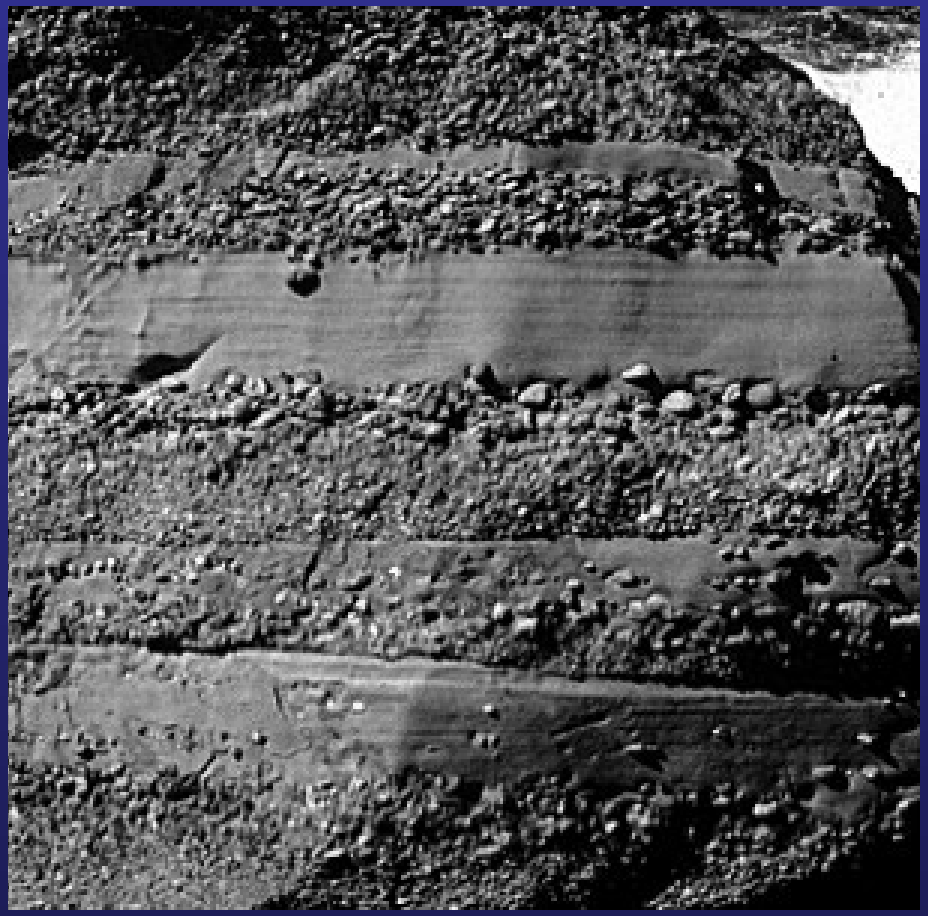


Conglomerate



Modern and ancient rivers laden with sediments
(as often found in mountains)

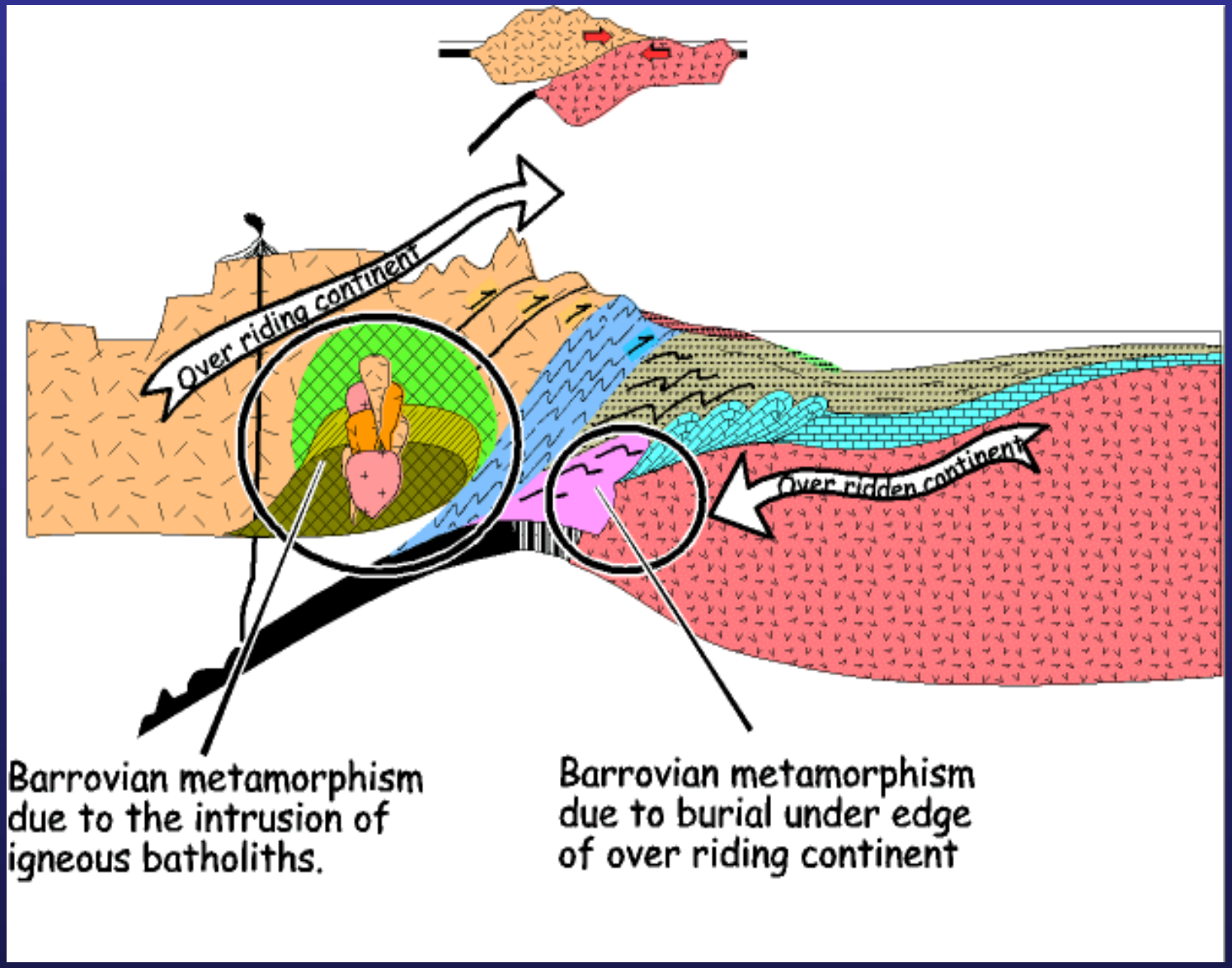
Grain size = Water Motion



Faster water can transport larger grains, leaving behind only the largest. Slowly moving water can only transport the finest particles.

The grain size of each sedimentary layer reveals whether this was a river or a lake at that particular time.

Metamorphic rocks: Regional metamorphism

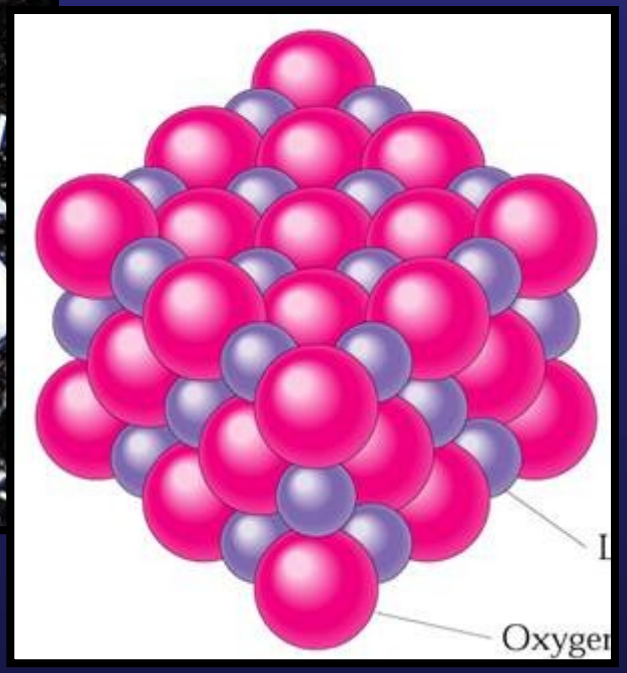


Rocks, minerals, and atoms





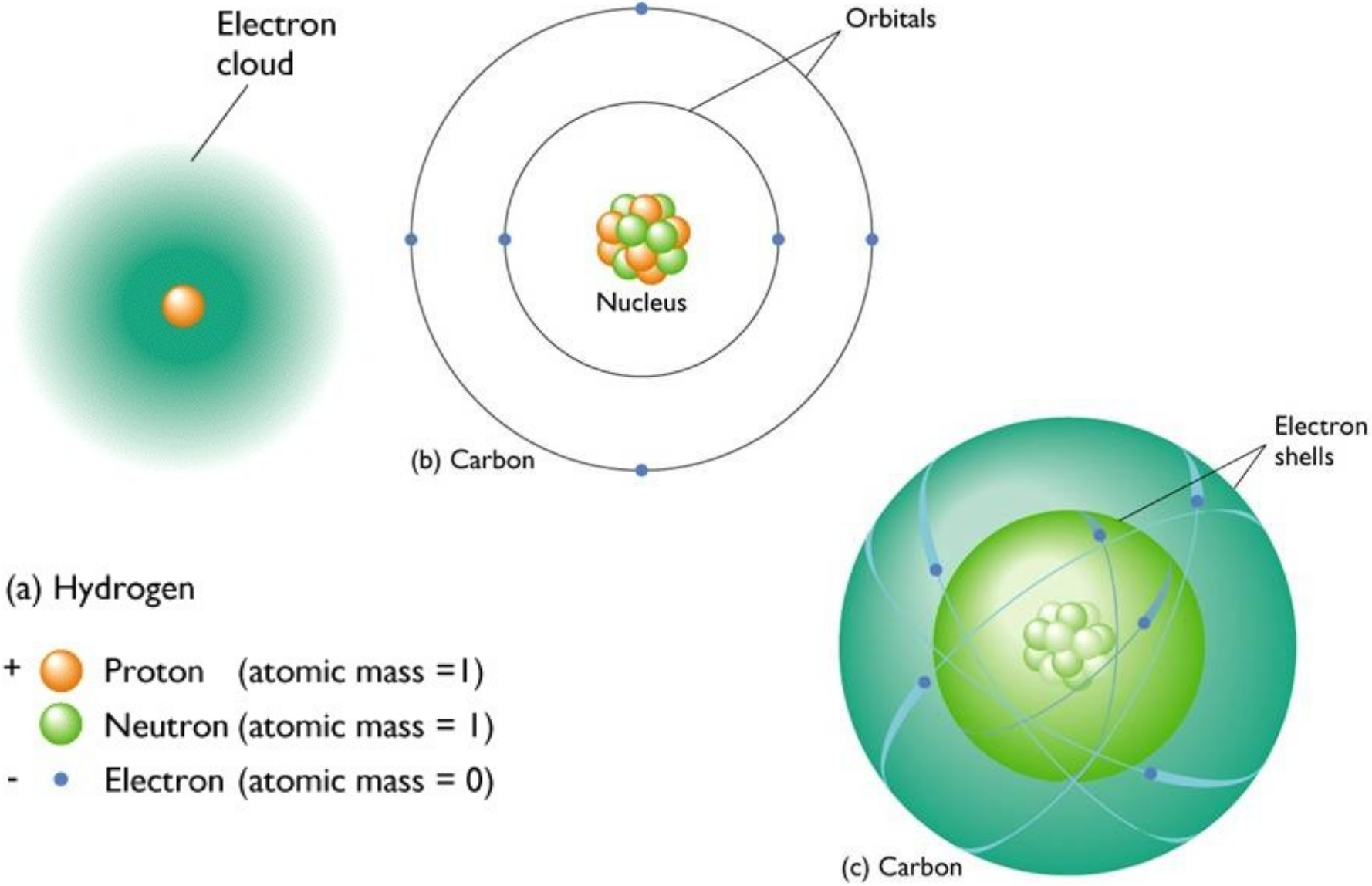
Rocks are made of minerals
are made of Atoms



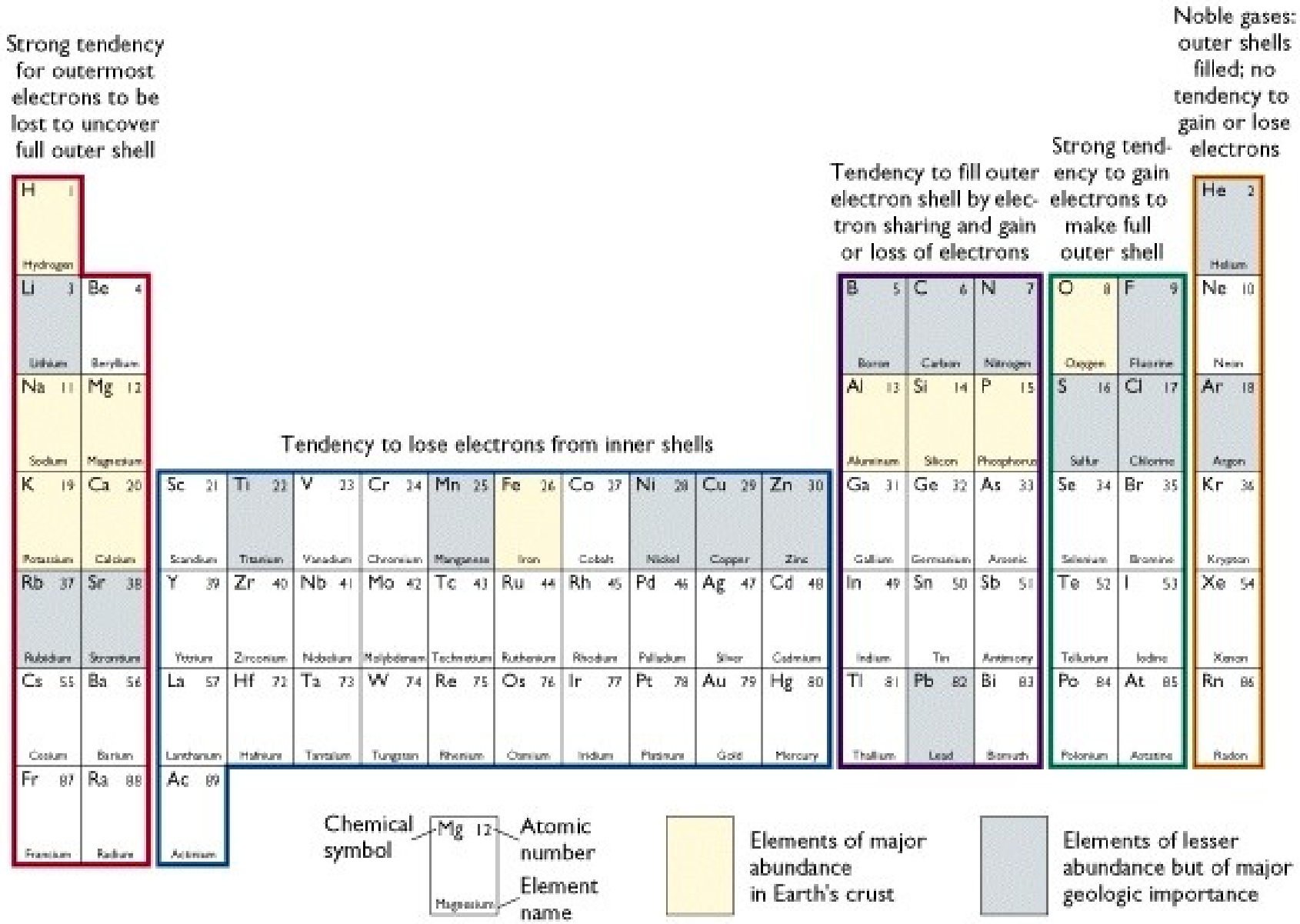
Pyrite (FeS_2) crystals



Atoms and their components



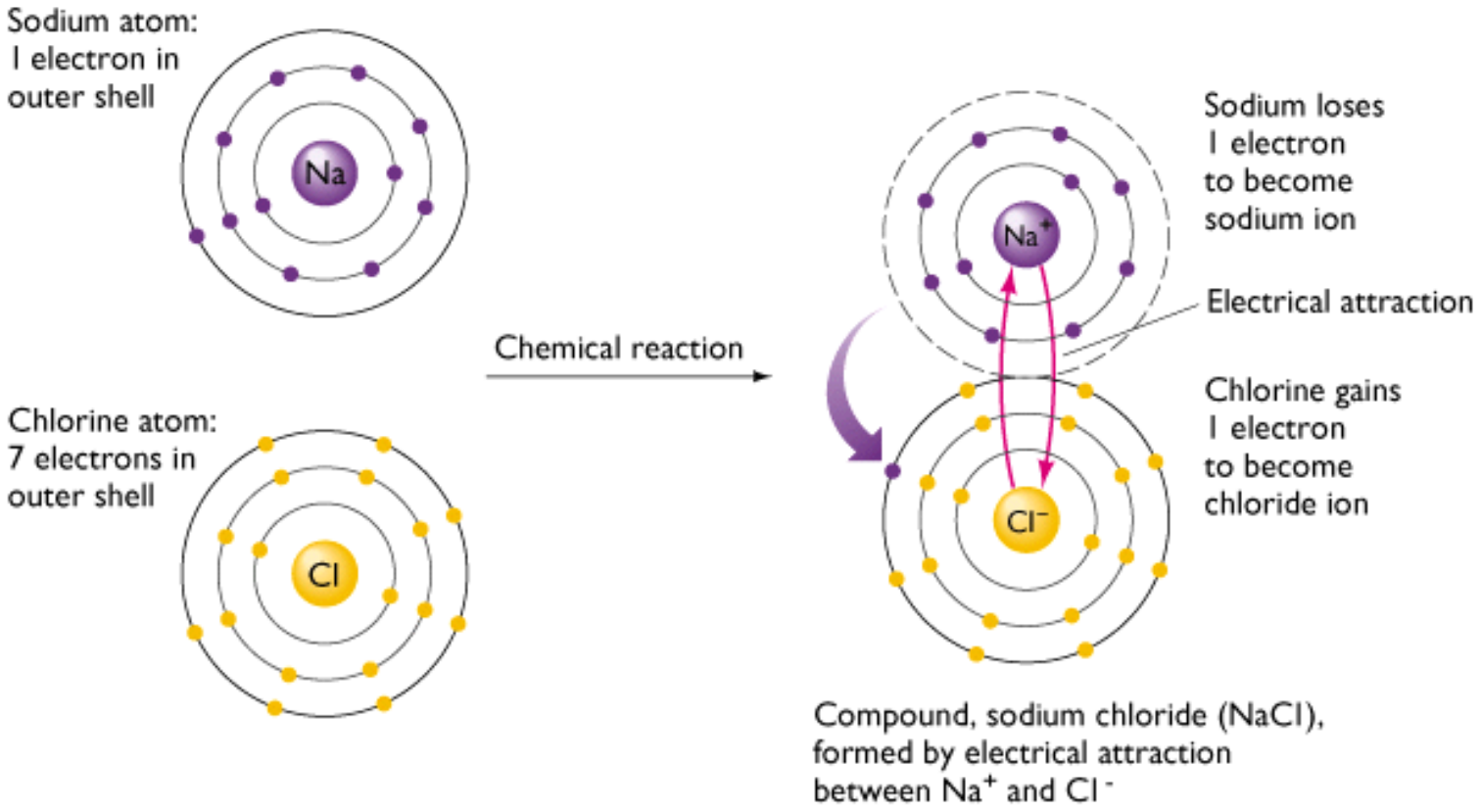
The Periodic Table: How atoms work



Electron exchange: ionic bonding

For neutrality, the positive charge of **protons** must be balanced by the negative charge of **electrons**

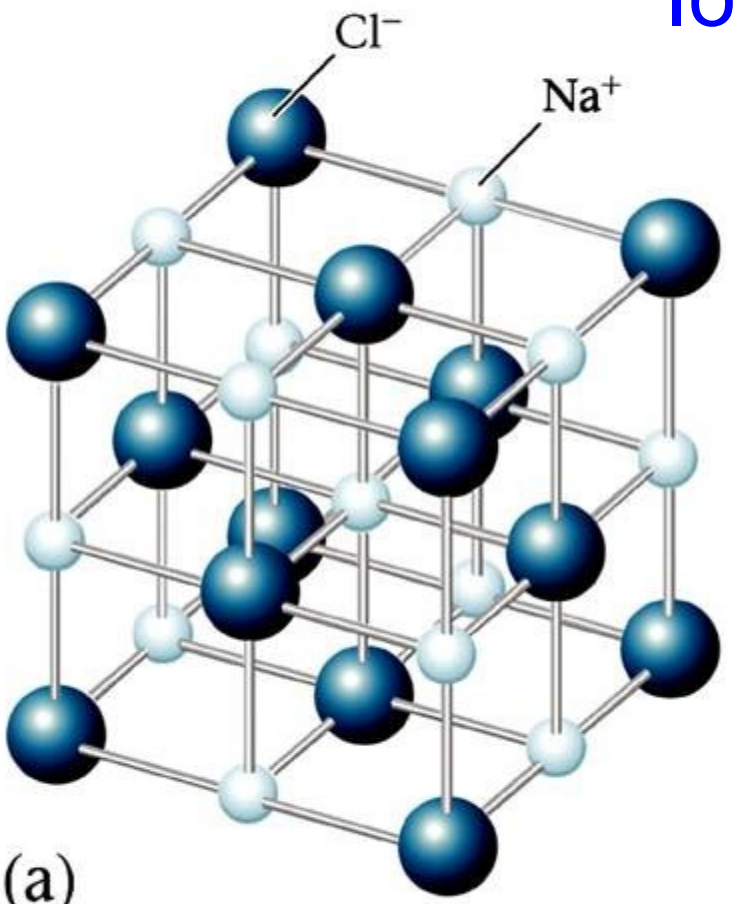
- **ion** -- a positive or negative charged atom

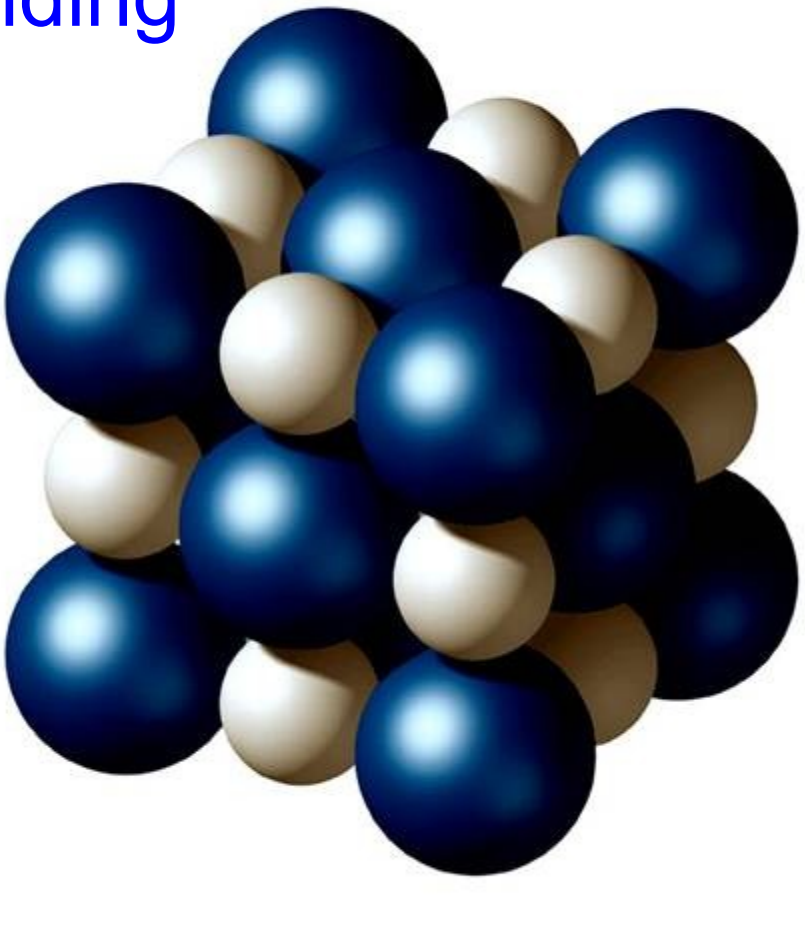


ionic bonding -- by gaining or losing electrons atoms become ions, which are attracted to each other by ionic bonds into molecules (smallest unit with the property of a compound)

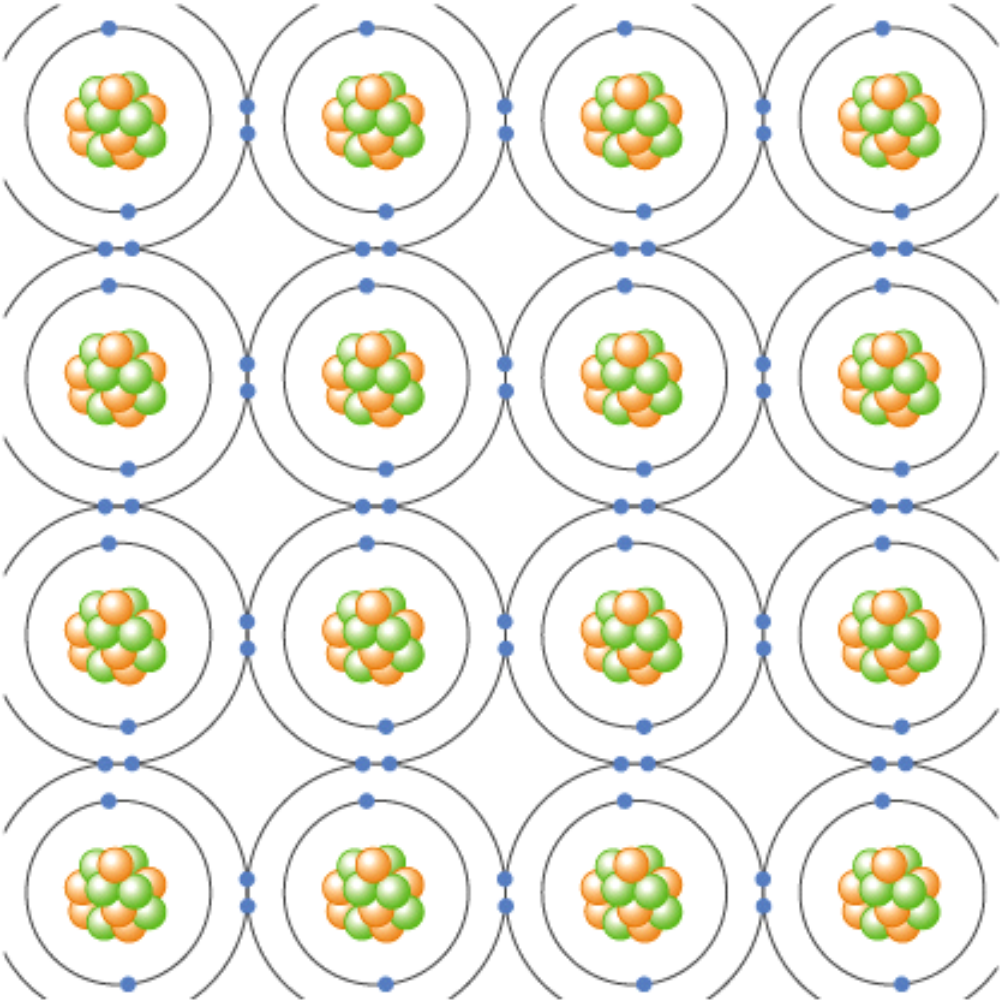
The atomic structure of halite (NaCl)

Ionic bonding



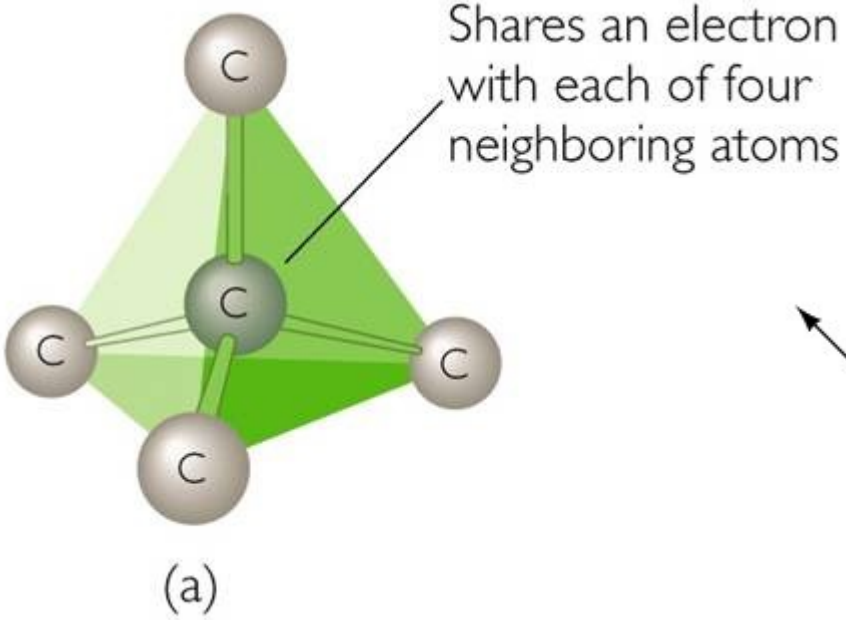


Electron sharing: covalent bonding

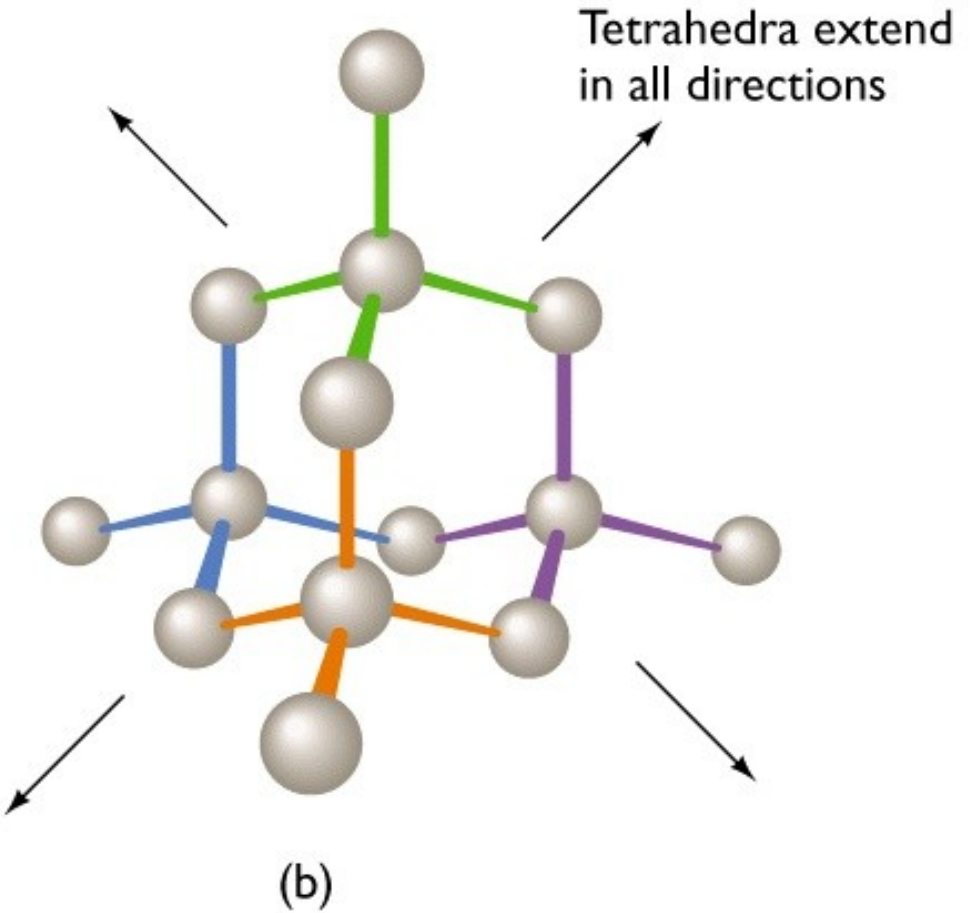


Diamond (made of carbon atoms)

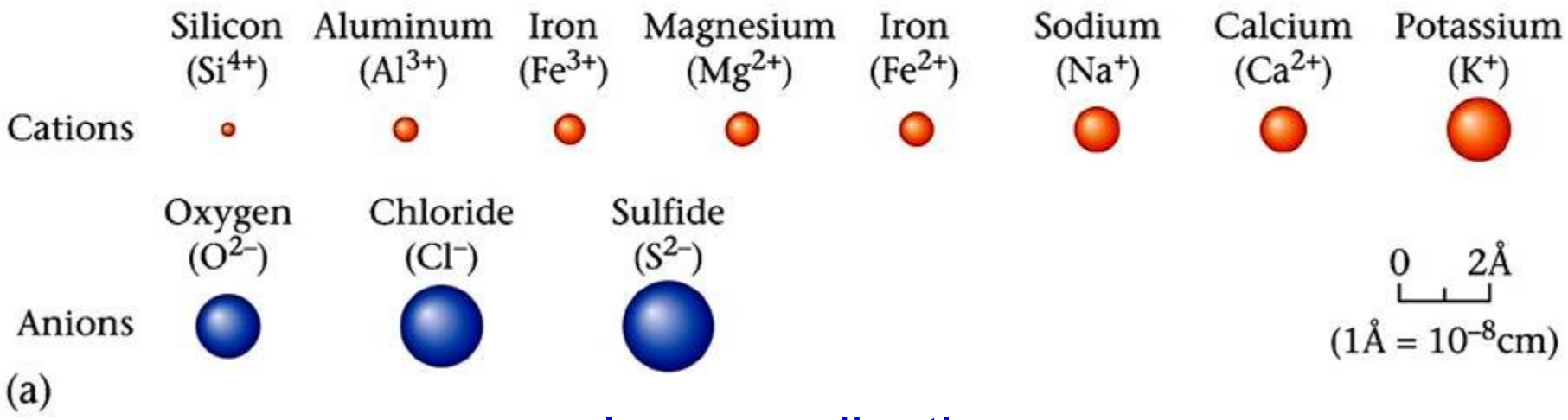
Covalent carbon bonds in diamond



Carbon is the element
Diamond is the mineral

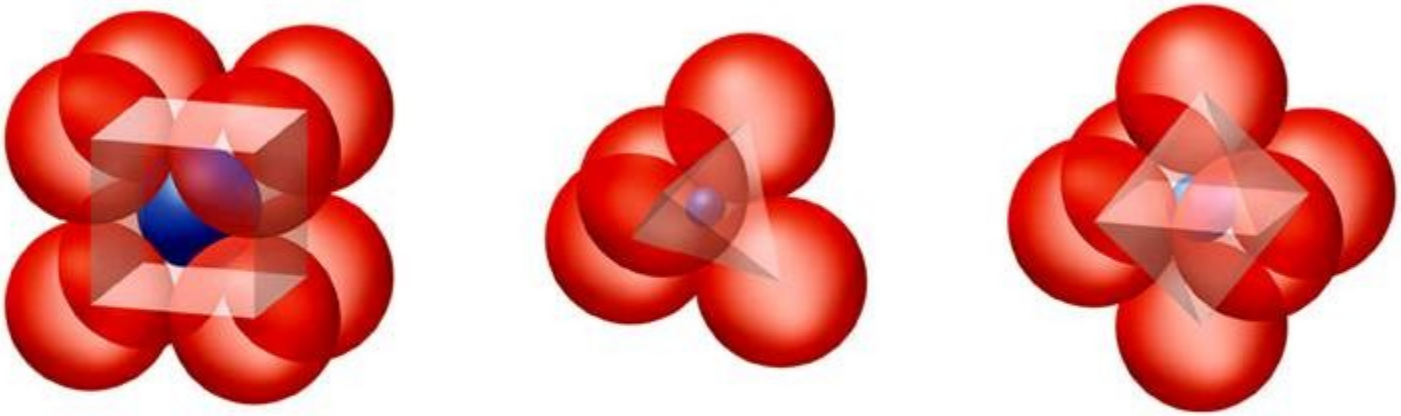


Relative sizes of ions



(a)

Ion co-ordination



(b) (c) (d)

What is a mineral?

The fundamental building block of rocks and soils . . .

a naturally occurring crystalline solid made of atoms containing a fixed or limited range of compositions with a systematic three-dimensional order. Its chemistry and physical properties are either fixed or fall within a limited range.

Importantly note that two minerals can be composed of the same element or elements, but because they have different structures they are different minerals.

Feldspars – two of the most important rock-forming minerals



Colours and forms of quartz





Quartz crystal displaying prism and pyramid faces

Quartz Crystal

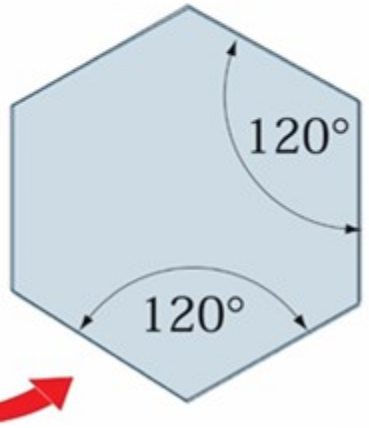
Faces display consistent angular relationships

Angles measured with a goniometer

Pyramid

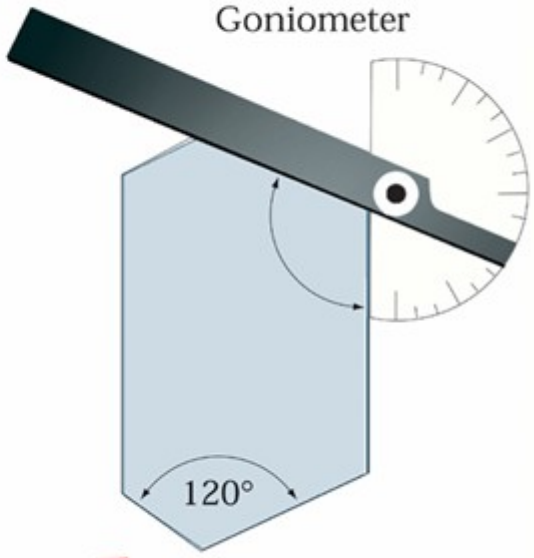


Basal cross section



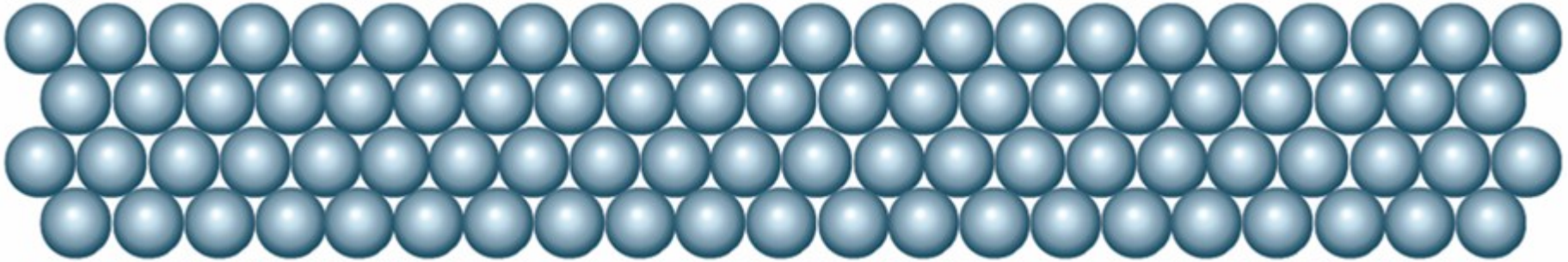
Prism



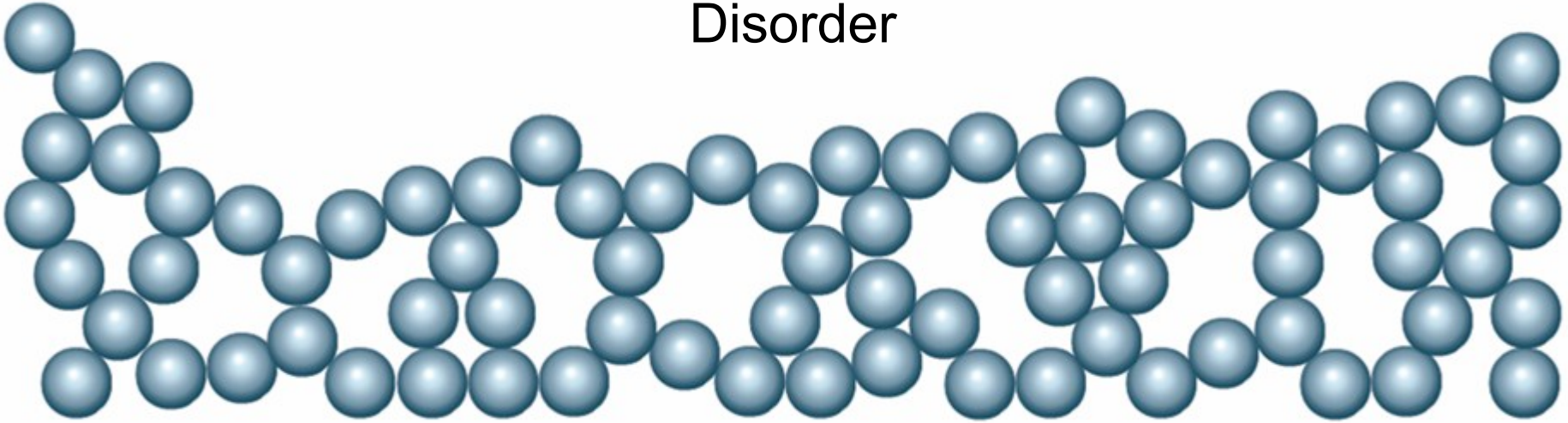


Crystals have ordered arrangements of atoms

Order



Disorder



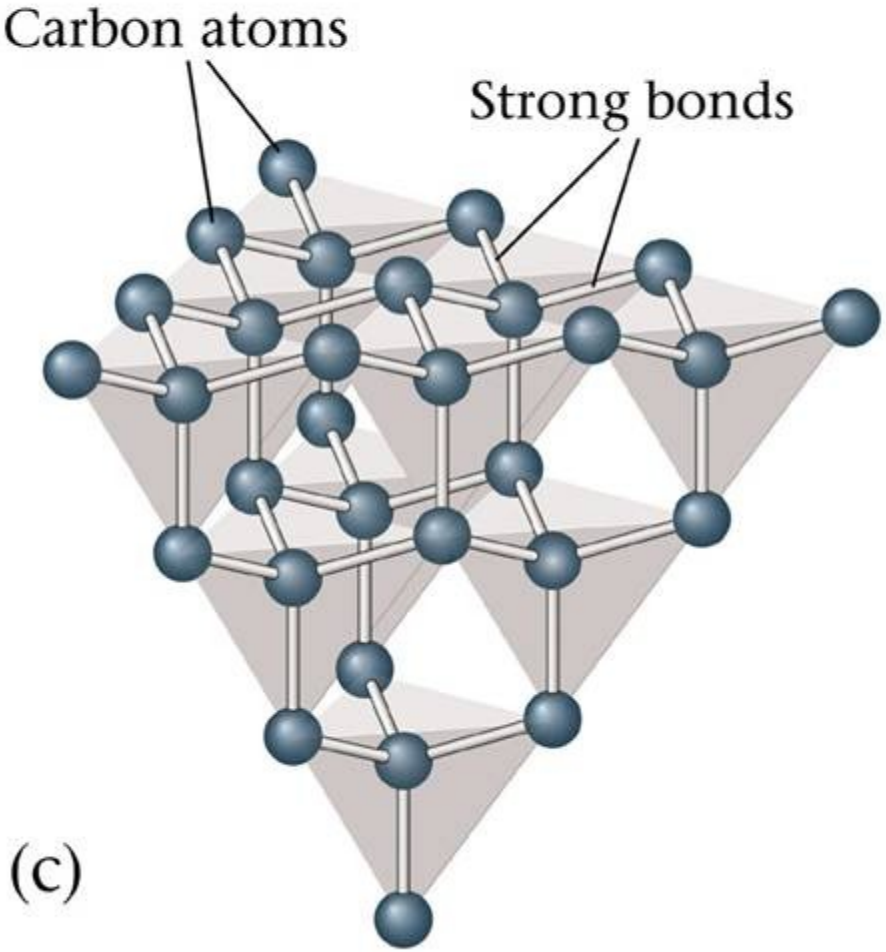
Minerals to know

Quartz	SiO_2
Alkali Feldspar	$(K,Na)AlSi_3O_8$
Mica	a sheet silicate
Olivine	$(Mg, Fe)_2SiO_4$ (most of the upper mantle)
Calcite	$CaCO_3$ (a rock made of this is limestone)
Pyrite	FeS_2 (fool's gold)
Graphite	C
Diamond	C
Halite	NaCl (table salt)
Magnetite	Fe_3O_4
Hematite	Fe_2O_3
Corundum	Al_2O_3 (gemstones are Rubies and Sapphires)
Apatite	Ca phosphate with OH, F, or Cl (bones and teeth)

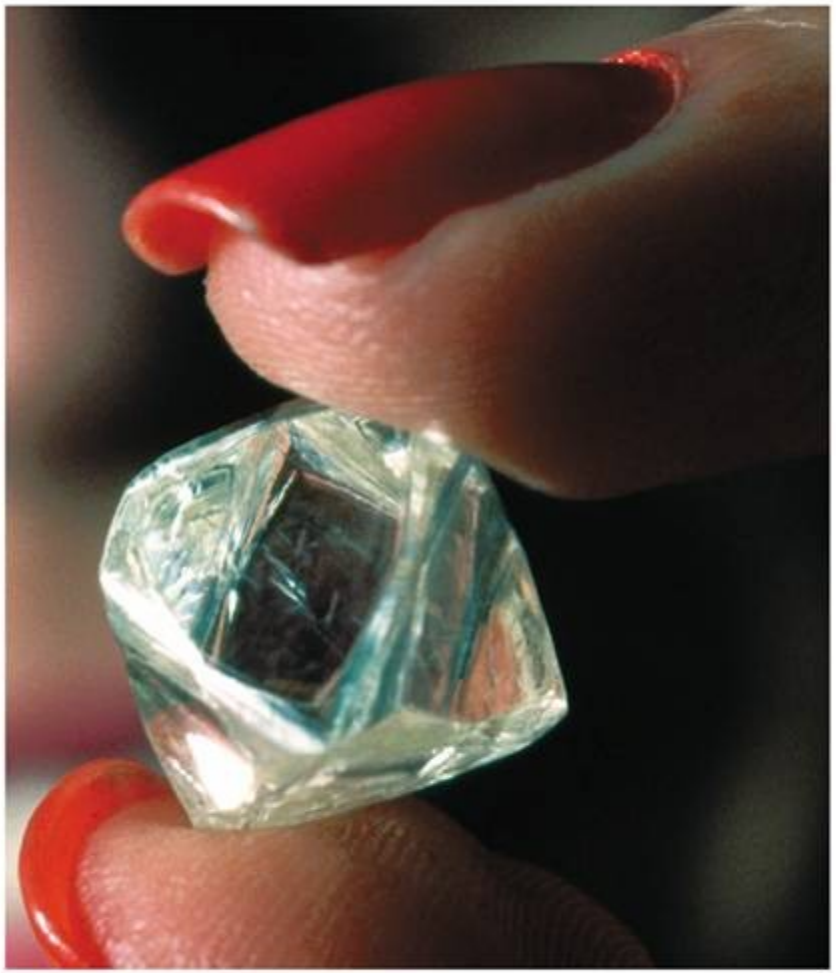
Bolded samples are passed around in class today, (+ coal)

The atomic structure of diamond

Covalent bonding

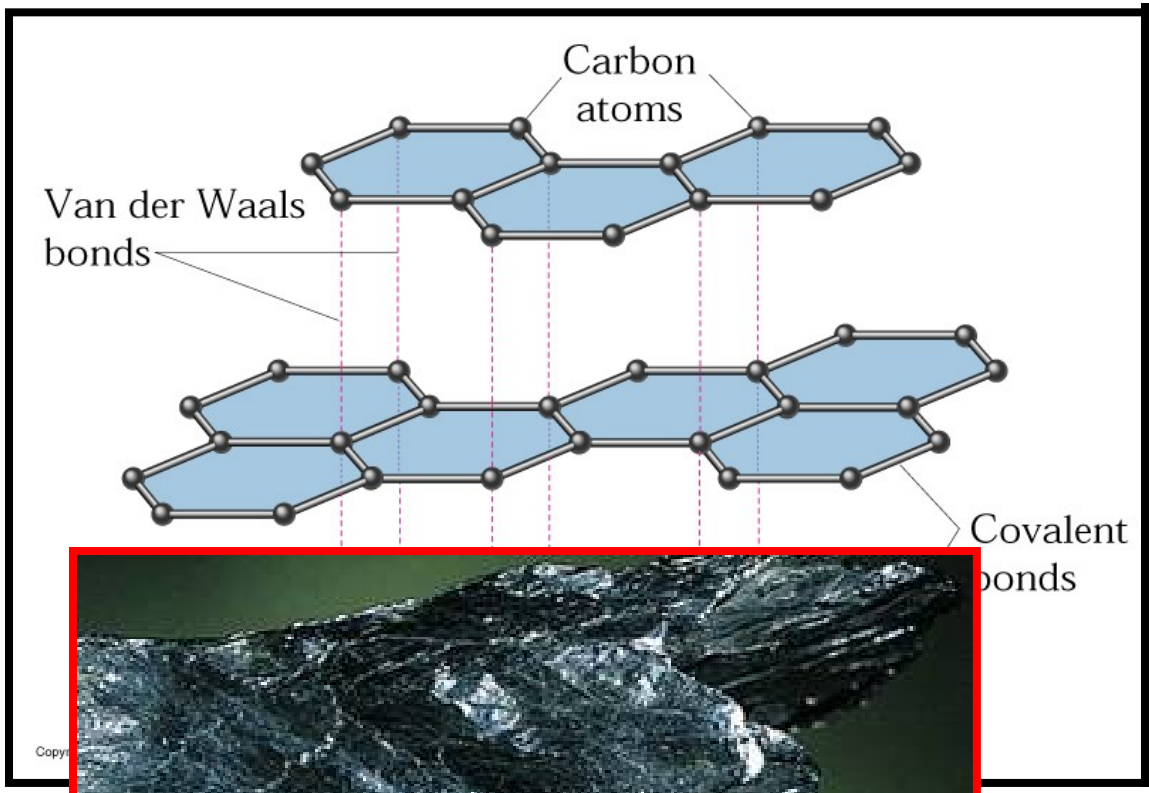


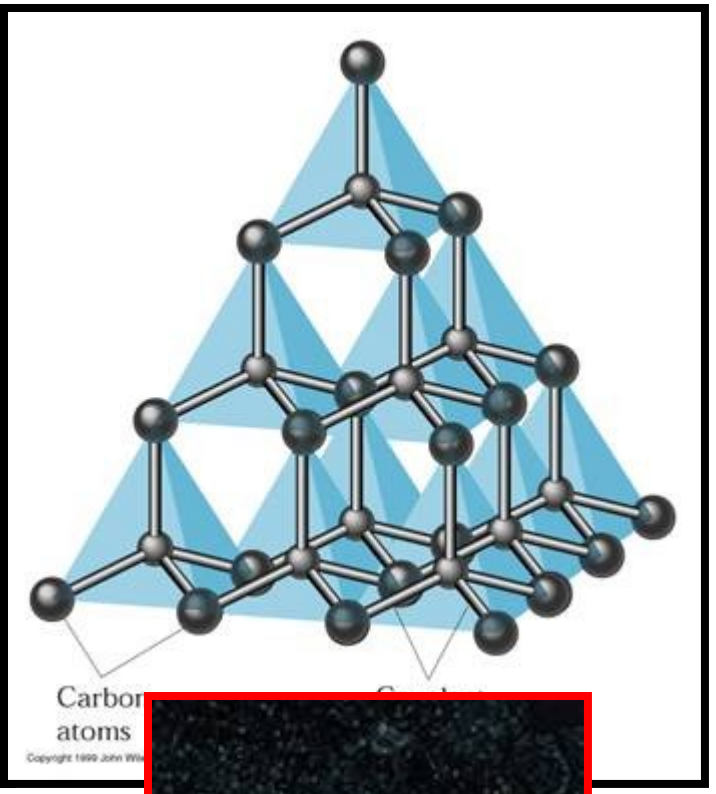
(c)



(d)

Same composition, different mineral









Why are minerals important?

GRAPHITE (NATURAL)

(Data in metric tons unless otherwise noted)

Domestic Production and Use: In 2021, natural graphite was not produced in the United States; however, approximately 95 U.S. companies, primarily in the Great Lakes and Northeastern regions and Alabama and Tennessee, consumed 45,000 tons valued at an estimated $41 million. The major uses of natural graphite were batteries, brake linings, lubricants, powdered metals, refractory applications, and steelmaking. During 2021, U.S. natural graphite imports were an estimated 53,000 tons, which were about 57% flake and high-purity, 42% amorphous, and 1% lump and chip graphite.

Salient Statistics—United States:	2017	2018	2019	2020	2021[e]
Production, mine	—	—	—	—	—
Imports for consumption	51,900	70,700	50,300	36,000	53,000
Exports	13,900	9,950	5,890	5,930	8,400
Consumption, apparent[1]	38,000	60,700	44,400	30,000	45,000
Price, average value of imports, dollars per ton at foreign ports:					
Flake	1,390	1,520	1,350	1,540	1,600
Lump and chip (Sri Lanka)	1,900	1,890	2,380	2,940	2,700
Amorphous	451	319	498	687	630
Net import reliance[1] as a percentage of apparent consumption	100	100	100	100	100

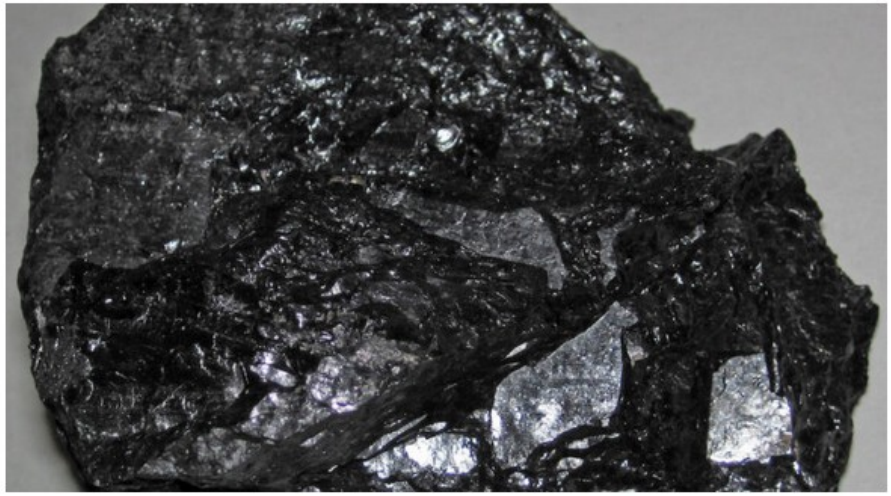
An essential part of lithium batteries

A Strategic Mineral: North America is dependent on overseas sources.

Why are minerals important?

Scientists one step closer to turning coal into graphite

Staff Writer | January 8, 2023 | 7:49 am Battery Metals Energy Suppliers & Equipment USA Coal Graphite



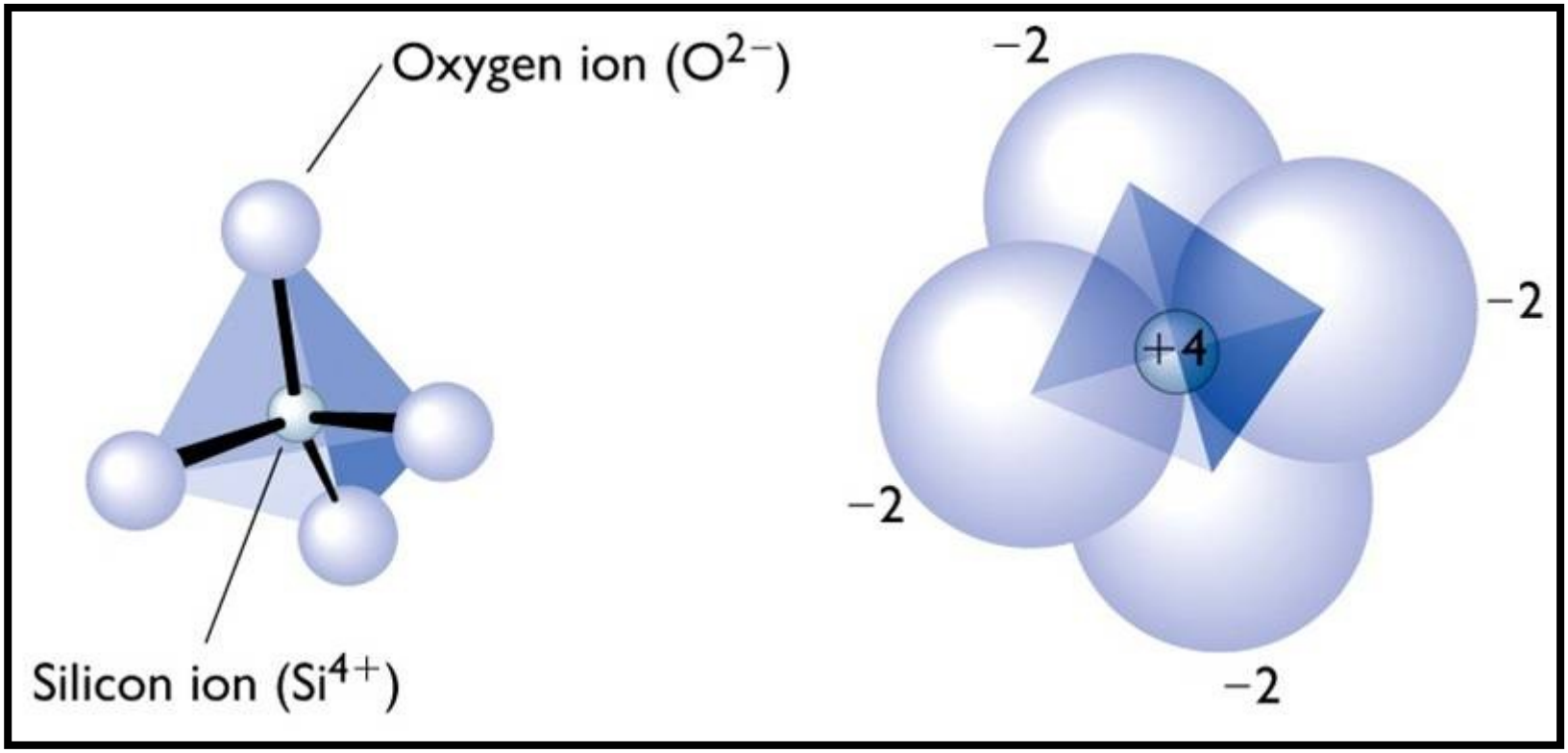
Coal. *(Reference image by James St. John, Flickr.)*

A team at Ohio University carried out a series of simulations showing how coal can be converted to valuable—and carbon-neutral—materials like graphite and carbon nanotubes.

Using the Pittsburgh Supercomputing Center's Bridges-2 system, the researchers simulated coal and graphite in computer software and recreated the coal-to-graphite conversion virtually. Generations of scientists know that, at least in theory, it is possible to convert coal to graphite if the fossil fuel is put under enough pressure at a high enough temperature.

SIGN UP FOR THE BATTE METALS DIGEST
SIGN UP

Silicate Minerals



Silicon and oxygen make up most rocks

- 45% of whole earth
- 75% of crust
- These rocks are called **silicates**

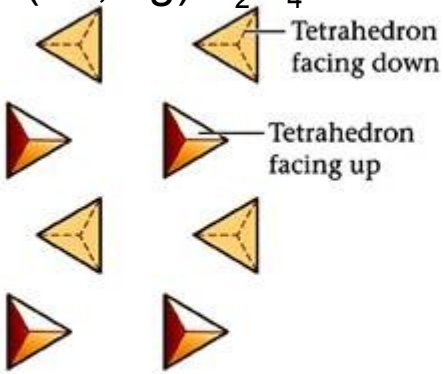
Very strong covalent bonds

- great material strength
- difficult to weather

Silicate Structures

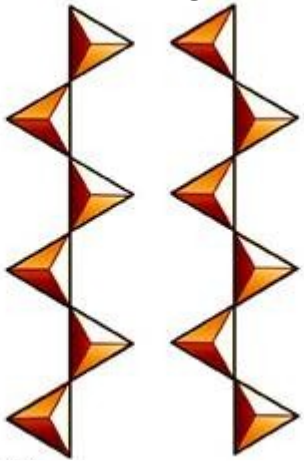
Olivine

$(Fe,Mg)Si_2O_4$



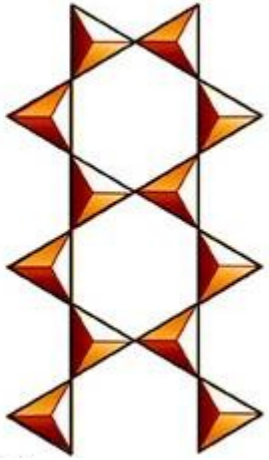
Pyroxene

$(Fe,Mg)SiO_3$

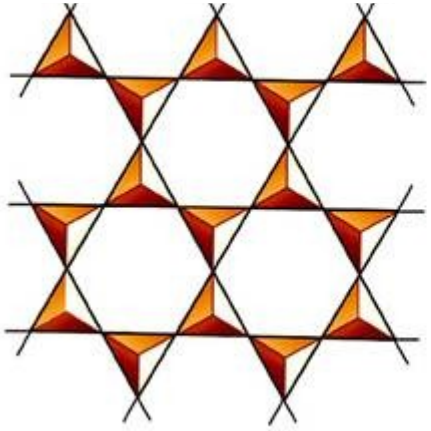


Amphibole

$(Ca,Mg,Fe)_7Si_8O_{22}(OH)_2$

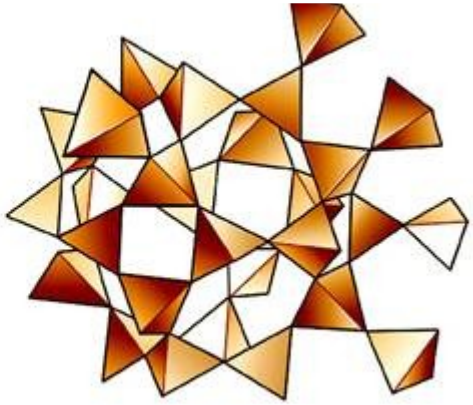


Mica



$K(Al,Mg,Fe)_3Si_3O_{10}(OH)_2$

Quartz, Feldspar



SiO_2, $(K,Na)AlSi_3O_8$

Why are minerals important?

Minerals found in rocks and soils are the hosts of elements and molecules. Some of these elements and molecules stored in minerals can significantly affect other parts of the Earth system.

Calcite, $CaCO_3$: stores most of Earth's CO_2

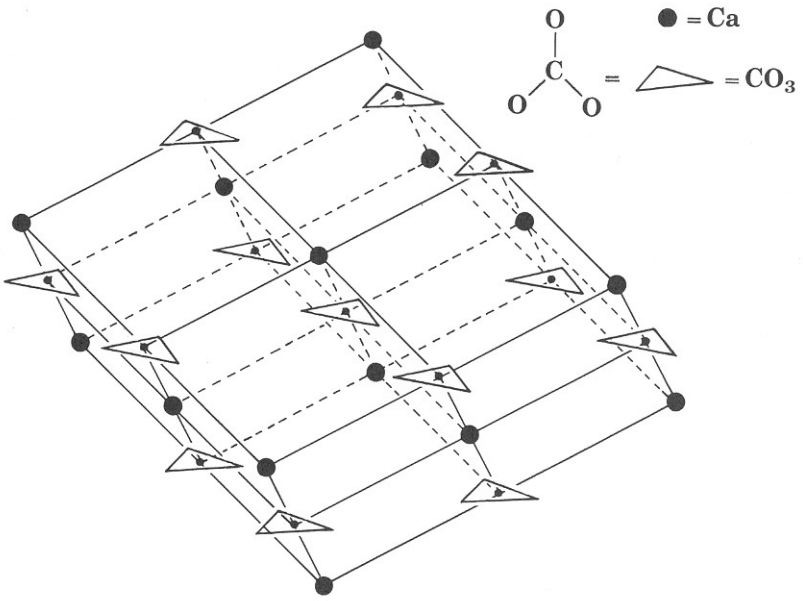


Fig. 9-8 Structure of calcite ($CaCO_3$). Note the distorted sodium chloride structure.



Chemical Weathering: Hydrolysis

Weak acids strip cations out of minerals

CO_2 + H_2O → H_2CO_3

Carbon dioxide + water → carbonic acid

$2KAlSi_3O_8$ + $2H_2CO_3$ + H_2O →

feldspar + carbonic acid + water

$Al_2Si_2O_5(OH)_4$ + $4SiO_2$ + $2K^+$ + $2\ HCO_3^-$

kaolinite (a clay) + silica (dissolved) + potassium (dissolved) + bicarbonate (dissolved)

These reactions requires water to be Present: either as liquid or as vapor

Crystallizing Minerals 'Unmix' solutions

Seawater solutes

Element	Concentration (by mass)
Chlorine	1.94%
Sodium	1.08%
Magnesium	0.129%
Sulfur	0.091%
Calcium	0.04%
Potassium	0.04%
Bromine	0.0067%
Carbon	0.0028%

(mostly carbonate and bicarbonate)

Seawater as an example

Sea Ice:
almost pure water (H_2O)

NCAR



Mineral Properties: Cleavage





Tendency of mineral to break along flat, planar surfaces

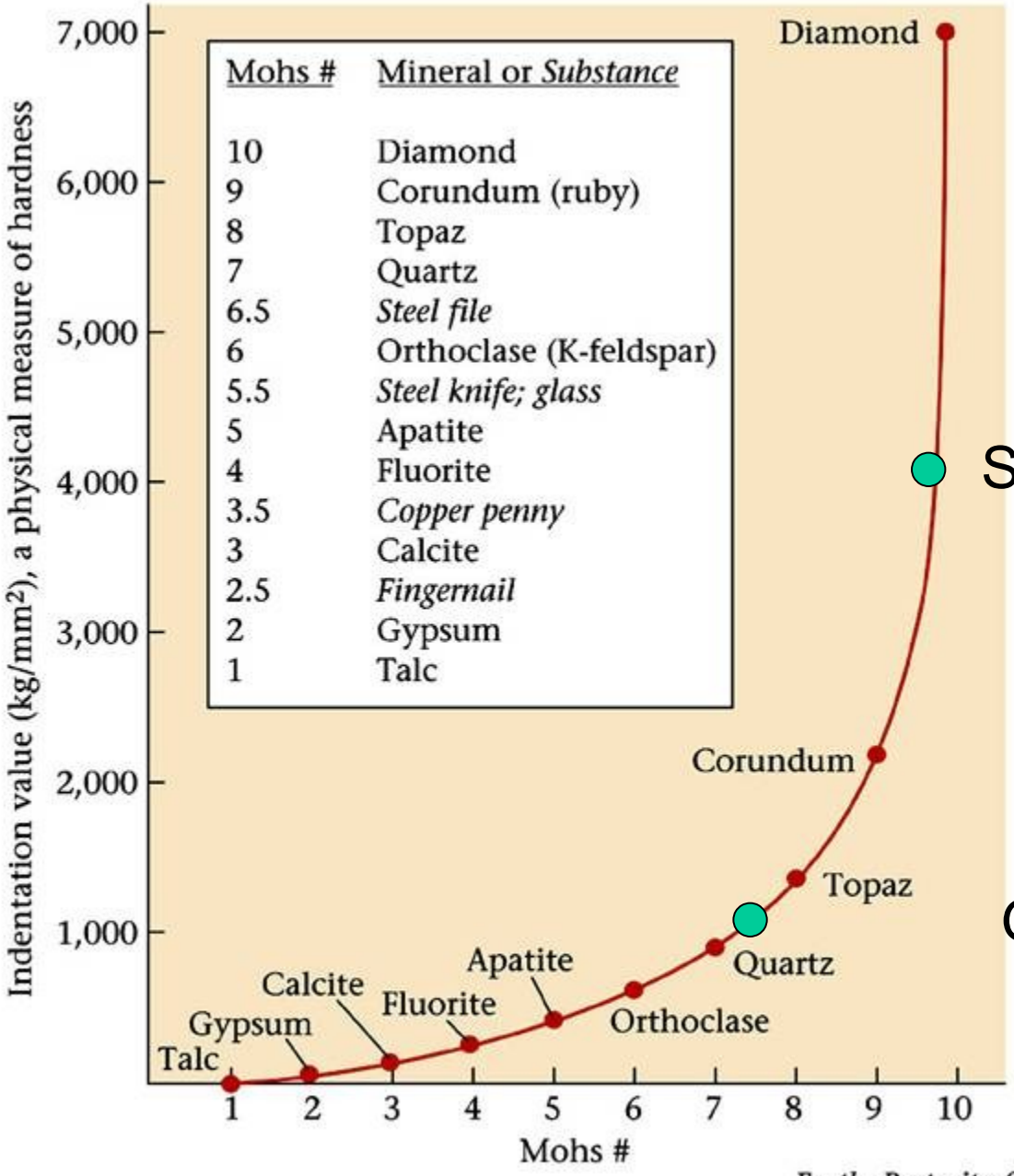
Governed by crystal structure

Mineral Properties: Hardness

Table 2.2 Mohs Scale of Hardness

Mineral	Scale Number	Common Objects
Talc	1	
Gypsum	2	— — — Fingernail
Calcite	3	— — — Copper coin
Fluorite	4	
Apatite	5	— — — Knife blade
Orthoclase	6	— — — Window glass
Quartz	7	— — — Steel file
Topaz	8	
Corundum	9	
Diamond	10	

Governed by crystal structure and chemistry



Silicon carbide

Garnet

TABLE 5.1

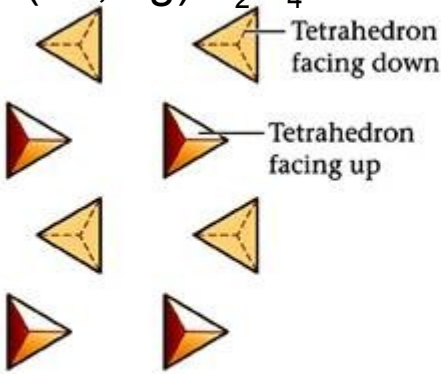
Earth: Portrait of a Planet, 2nd Edition

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Silicate Structures

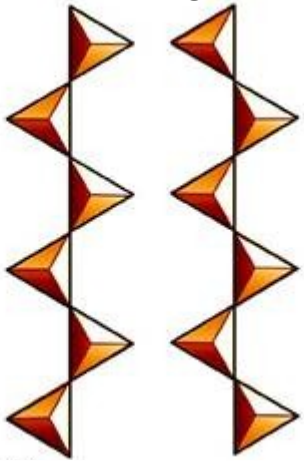
Olivine

$(Fe,Mg)Si_2O_4$



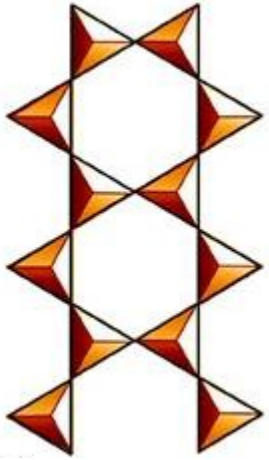
Pyroxene

$(Fe,Mg)SiO_3$

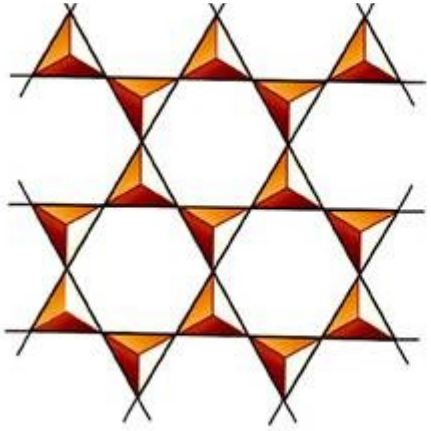


Amphibole

$(Ca,Mg,Fe)_7Si_8O_{22}(OH)_2$

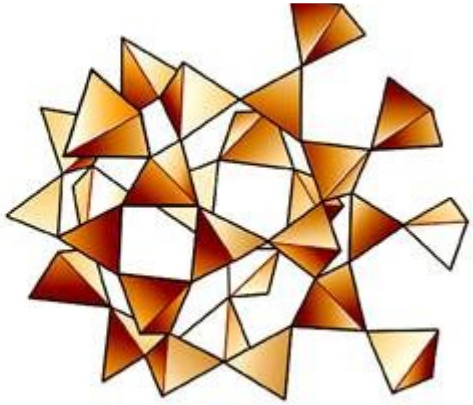


Mica



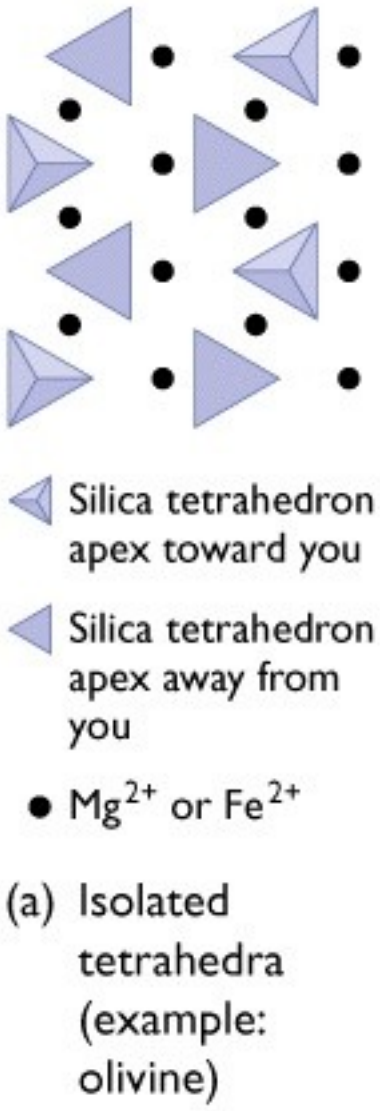
$K(Al,Mg,Fe)_3Si_3O_{10}(OH)_2$

Quartz, Feldspar



SiO_2, $(K,Na)AlSi_3O_8$

Isolated silicate tetrahedra



(a) Isolated tetrahedra (example: olivine)

- Simplest silicate structure
- Lots of metal cations (dense)

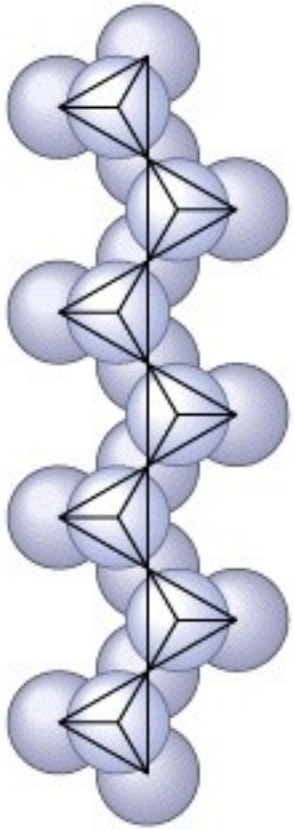


Olivine

$(Mg, Fe)_2SiO_4$

Chain Silicates

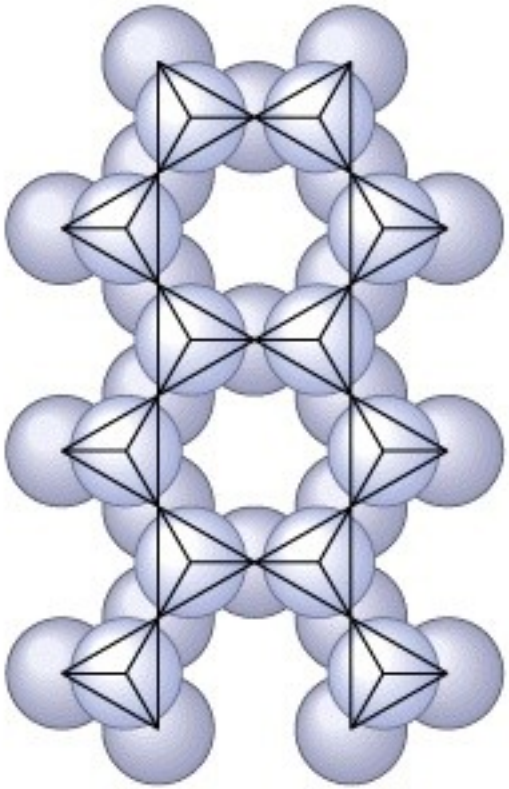
Chains make long, rod-like minerals



(b) Single chain (example: pyroxene)



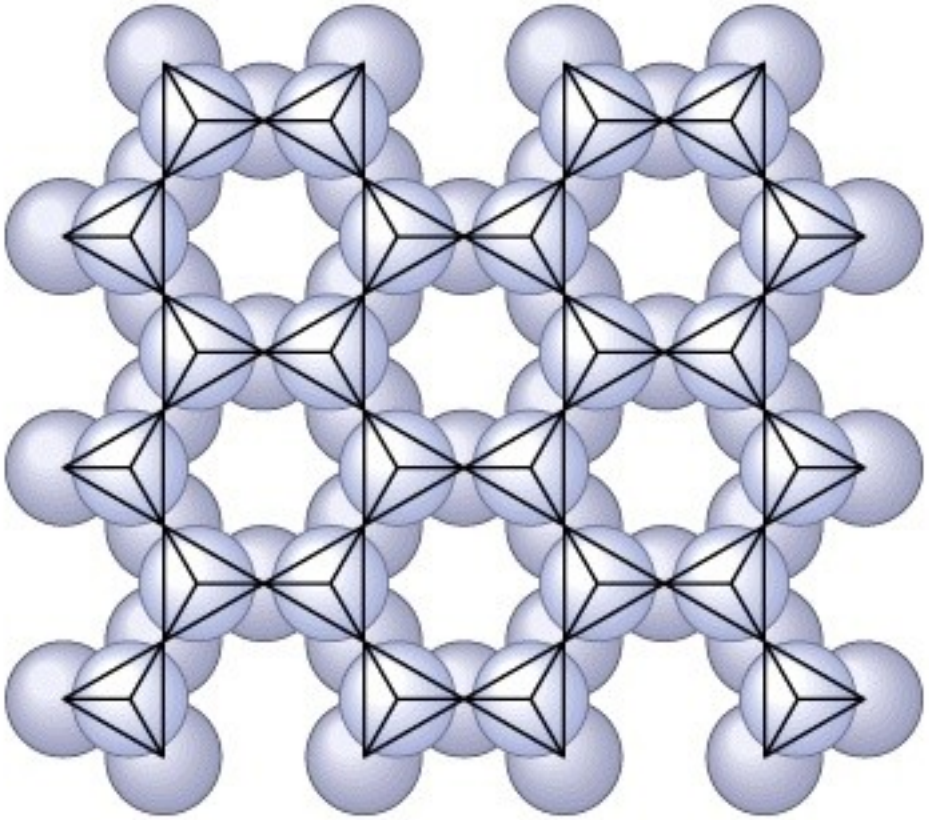




(c) Double chain (example: amphibole)

Sheet Silicates

Sheet-like minerals (includes micas and many clays)

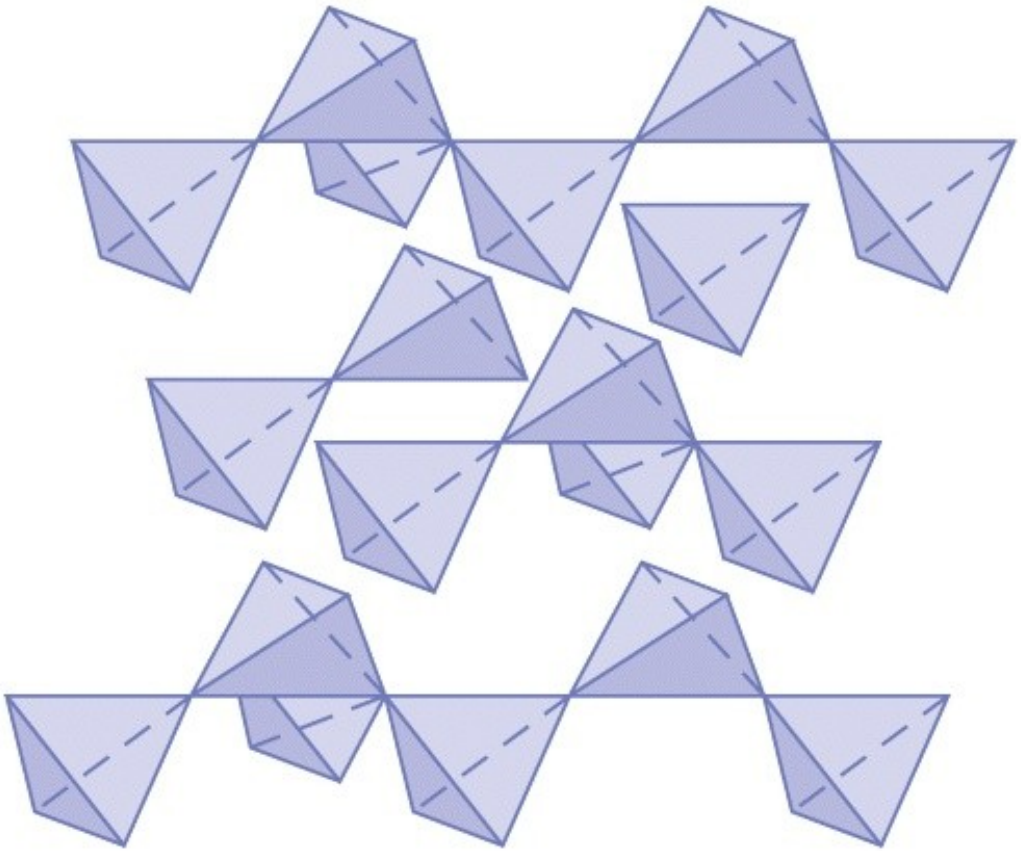


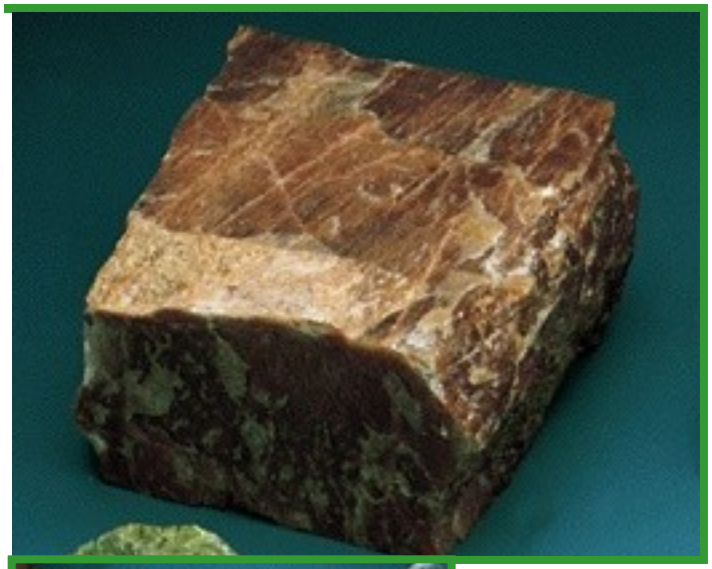
(d) Sheet (example: mica)



Framework Silicates

3-D lattice works of linked silica ions (includes quartz and feldspars)







Non-Silicate Minerals



Important mineral class for many reasons

- Carbonates: 1/5 of world's sedimentary rocks
- Oxides: Major ores
- Sulfides: Major ores
- Phosphates: Biostructures (teeth), fertilizer

They have unique mechanical and physical properties